ELEMENTS BOOK 5

Proportion[†]

[†]The theory of proportion set out in this book is generally attributed to Eudoxus of Cnidus. The novel feature of this theory is its ability to deal with irrational magnitudes, which had hitherto been a major stumbling block for Greek mathematicians. Throughout the footnotes in this book, α, β, γ, *etc.*, denote general (possibly irrational) magnitudes, whereas m, n, l, *etc.*, denote positive integers.

Ὅροι.

α′. Μέρος ἐστὶ μέγεθος μεγέθους τὸ ἔλασσον τοῦ μείζονος, ὅταν καταμετρῇ τὸ μεῖζον.

β′. Πολλαπλάσιον δὲ τὸ μεῖζον τοῦ ἐλάττονος, ὅταν καταμετρῆται ὑπὸ τοῦ ἐλάττονος.

γ′. Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἡ κατὰ πηλικότητά ποια σχέσις.

δ′. Λόγον ἔχειν πρὸς ἄλληλα μεγέθη λέγεται, ἃ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέχειν.

ε′. Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρὸς δεύτερον καὶ τρίτον πρὸς τέταρτον, ὅταν τὰ τοῦ πρώτου καί τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ' ὁποιονοῦν πολλαπλασιασμὸν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέχῃ ἢ ἅμα ἴσα ᾖ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.

ϛ′. Τὰ δὲ τὸν αὐτὸν ἔχοντα λόγον μεγέθη ἀνάλογον καλείσθω.

ζ′. Ὅταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέχῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέχῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρὸς τὸ δεύτερον μείζονα λόγον ἔχειν λέγεται, ἤπερ τὸ τρίτον πρὸς τὸ τέταρτον.

η′. Ἀναλογία δὲ ἐν τρισὶν ὅροις ἐλαχίστη ἐστίν.

θ′. Ὅταν δὲ τρία μεγέθη ἀνάλογον ᾖ, τὸ πρῶτον πρὸς τὸ τρίτον διπλασίονα λόγον ἔχειν λέγεται ἤπερ πρὸς τὸ δεύτερον.

ι′. Ὅταν δὲ τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔχειν λέγεται ἤπερ πρὸς τὸ δεύτερον, καὶ ἀεὶ ἑξῆς ὁμοίως, ὡς ἂν ἡ ἀναλογία ὑπάρχῃ.

ια′. Ὁμόλογα μεγέθη λέγεται τὰ μὲν ἡγούμενα τοῖς ἡγουμένοις τὰ δὲ ἑπόμενα τοῖς ἑπομένοις.

ιβ′. Ἐναλλὰξ λόγος ἐστὶ λῆψις τοῦ ἡγουμένου πρὸς τὸ ἡγούμενον καὶ τοῦ ἑπομένου πρὸς τὸ ἑπόμενον.

ιγ′. Ἀνάπαλιν λόγος ἐστὶ λῆψις τοῦ ἑπομένου ὡς ἡγουμένου πρὸς τὸ ἡγούμενον ὡς ἑπόμενον.

ιδ′. Σύνθεσις λόγου ἐστὶ λῆψις τοῦ ἡγουμένου μετὰ τοῦ ἑπομένου ὡς ἑνὸς πρὸς αὐτὸ τὸ ἑπόμενον.

ιε′. Διαίρεσις λόγου ἐστὶ λῆψις τῆς ὑπεροχῆς, ᾗ ὑπερέχει τὸ ἡγούμενον τοῦ ἑπομένου, πρὸς αὐτὸ τὸ ἑπόμενον.

ιϛ′. Ἀναστροφὴ λόγου ἐστὶ λῆψις τοῦ ἡγουμένου πρὸς τὴν ὑπεροχήν, ᾗ ὑπερέχει τὸ ἡγούμενον τοῦ ἑπομένου.

ιζ′. Δι' ἴσου λόγος ἐστὶ πλειόνων ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος σύνδυο λαμβανομένων καὶ ἐν τῷ αὐτῷ λόγῳ, ὅταν ᾖ ὡς ἐν τοῖς πρώτοις μεγέθεσι τὸ πρῶτον πρὸς τὸ ἔσχατον, οὕτως ἐν τοῖς δευτέροις μεγέθεσι τὸ πρῶτον πρὸς τὸ ἔσχατον· ἢ ἄλλως· λῆψις τῶν ἄκρων

Definitions

1. A magnitude is a part of a(nother) magnitude, the lesser of the greater, when it measures the greater.†

2. And the greater (magnitude is) a multiple of the lesser when it is measured by the lesser.

3. A ratio is a certain type of condition with respect to size of two magnitudes of the same kind.‡

4. (Those) magnitudes are said to have a ratio with respect to one another which, being multiplied, are capable of exceeding one another.§

5. Magnitudes are said to be in the same ratio, the first to the second, and the third to the fourth, when equal multiples of the first and the third either both exceed, are both equal to, or are both less than, equal multiples of the second and the fourth, respectively, being taken in corresponding order, according to any kind of multiplication whatever.¶

6. And let magnitudes having the same ratio be called proportional.*

7. And when for equal multiples (as in Def. 5), the multiple of the first (magnitude) exceeds the multiple of the second, and the multiple of the third (magnitude) does not exceed the multiple of the fourth, then the first (magnitude) is said to have a greater ratio to the second than the third (magnitude has) to the fourth.

8. And a proportion in three terms is the smallest (possible).$

9. And when three magnitudes are proportional, the first is said to have to the third the squared‖ ratio of that (it has) to the second.††

10. And when four magnitudes are (continuously) proportional, the first is said to have to the fourth the cubed‡‡ ratio of that (it has) to the second.§§ And so on, similarly, in successive order, whatever the (continuous) proportion might be.

11. These magnitudes are said to be corresponding (magnitudes): the leading to the leading (of two ratios), and the following to the following.

12. An alternate ratio is a taking of the (ratio of the) leading (magnitude) to the leading (of two equal ratios), and (setting it equal to) the (ratio of the) following (magnitude) to the following.¶¶

13. An inverse ratio is a taking of the (ratio of the) following (magnitude) as the leading and the leading (magnitude) as the following.**

14. A composition of a ratio is a taking of the (ratio of the) leading plus the following (magnitudes), as one, to the following (magnitude) by itself.$$

καθ᾽ ὑπεξαίρεσιν τῶν μέσων.

ιη′. Τεταραγμένη δὲ ἀναλογία ἐστίν, ὅταν τριῶν ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος γίνηται ὡς μὲν ἐν τοῖς πρώτοις μεγέθεσιν ἡγούμενον πρὸς ἑπόμενον, οὕτως ἐν τοῖς δευτέροις μεγέθεσιν ἡγούμενον πρὸς ἑπόμενον, ὡς δὲ ἐν τοῖς πρώτοις μεγέθεσιν ἑπόμενον πρὸς ἄλλο τι, οὕτως ἐν τοῖς δευτέροις ἄλλο τι πρὸς ἡγούμενον.

15. A separation of a ratio is a taking of the (ratio of the) excess by which the leading (magnitude) exceeds the following to the following (magnitude) by itself.‖‖‖

16. A conversion of a ratio is a taking of the (ratio of the) leading (magnitude) to the excess by which the leading (magnitude) exceeds the following.†††

17. There being several magnitudes, and other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, a ratio via equality (or *ex aequali*) occurs when as the first is to the last in the first (set of) magnitudes, so the first (is) to the last in the second (set of) magnitudes. Or alternately, (it is) a taking of the (ratio of the) outer (magnitudes) by the removal of the inner (magnitudes).‡‡‡

18. There being three magnitudes, and other (magnitudes) of equal number to them, a perturbed proportion occurs when as the leading is to the following in the first (set of) magnitudes, so the leading (is) to the following in the second (set of) magnitudes, and as the following (is) to some other (*i.e.*, the remaining magnitude) in the first (set of) magnitudes, so some other (is) to the leading in the second (set of) magnitudes.§§§

† In other words, α is said to be a part of β if $\beta = m\,\alpha$.

‡ In modern notation, the ratio of two magnitudes, α and β, is denoted $\alpha : \beta$.

§ In other words, α has a ratio with respect to β if $m\,\alpha > \beta$ and $n\,\beta > \alpha$, for some m and n.

¶ In other words, $\alpha : \beta :: \gamma : \delta$ if and only if $m\,\alpha > n\,\beta$ whenever $m\,\gamma > n\,\delta$, and $m\,\alpha = n\,\beta$ whenever $m\,\gamma = n\,\delta$, and $m\,\alpha < n\,\beta$ whenever $m\,\gamma < n\,\delta$, for all m and n. This definition is the kernel of Eudoxus' theory of proportion, and is valid even if α, β, *etc.*, are irrational.

* Thus if α and β have the same ratio as γ and δ then they are proportional. In modern notation, $\alpha : \beta :: \gamma : \delta$.

$ In modern notation, a proportion in three terms—α, β, and γ—is written: $\alpha : \beta :: \beta : \gamma$.

‖ Literally, "double".

†† In other words, if $\alpha : \beta :: \beta : \gamma$ then $\alpha : \gamma :: \alpha^2 : \beta^2$.

‡‡ Literally, "triple".

§§ In other words, if $\alpha : \beta :: \beta : \gamma :: \gamma : \delta$ then $\alpha : \delta :: \alpha^3 : \beta^3$.

¶¶ In other words, if $\alpha : \beta :: \gamma : \delta$ then the alternate ratio corresponds to $\alpha : \gamma :: \beta : \delta$.

** In other words, if $\alpha : \beta$ then the inverse ratio corresponds to $\beta : \alpha$.

$$ In other words, if $\alpha : \beta$ then the composed ratio corresponds to $\alpha + \beta : \beta$.

‖‖‖ In other words, if $\alpha : \beta$ then the separated ratio corresponds to $\alpha - \beta : \beta$.

††† In other words, if $\alpha : \beta$ then the converted ratio corresponds to $\alpha : \alpha - \beta$.

‡‡‡ In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta : \gamma :: \delta : \epsilon : \zeta$, then the ratio via equality (or *ex aequali*) corresponds to $\alpha : \gamma :: \delta : \zeta$.

§§§ In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta :: \delta : \epsilon$ as well as $\beta : \gamma :: \zeta : \delta$, then the proportion is said to be perturbed.

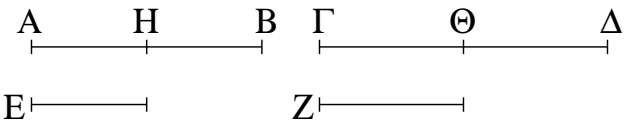
α′.

Proposition 1†

᾽Εὰν ᾖ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἕκαστον ἑκάστου ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἓν τῶν μεγεθῶν ἑνός, τοσαυταπλάσια ἔσται καὶ τὰ

If there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), then as many

πάντα τῶν πάντων.

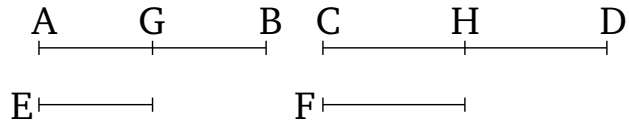


Ἔστω ὁποσαοῦν μεγέθη τὰ ΑΒ, ΓΔ ὁποσωνοῦν μεγεθῶν τῶν Ε, Ζ ἴσων τὸ πλῆθος ἕκαστον ἑκάστου ἰσάκις πολλαπλάσιον· λέγω, ὅτι ὁσαπλάσιόν ἐστι τὸ ΑΒ τοῦ Ε, τοσαυταπλάσια ἔσται καὶ τὰ ΑΒ, ΓΔ τῶν Ε, Ζ.

Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Ε καὶ τὸ ΓΔ τοῦ Ζ, ὅσα ἄρα ἐστὶν ἐν τῷ ΑΒ μεγέθη ἴσα τῷ Ε, τοσαῦτα καὶ ἐν τῷ ΓΔ ἴσα τῷ Ζ. διῃρήσθω τὸ μὲν ΑΒ εἰς τὰ τῷ Ε μεγέθη ἴσα τὰ ΑΗ, ΗΒ, τὸ δὲ ΓΔ εἰς τὰ τῷ Ζ ἴσα τὰ ΓΘ, ΘΔ· ἔσται δὴ ἴσον τὸ πλῆθος τῶν ΑΗ, ΗΒ τῷ πλήθει τῶν ΓΘ, ΘΔ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν ΑΗ τῷ Ε, τὸ δὲ ΓΘ τῷ Ζ, ἴσον ἄρα τὸ ΑΗ τῷ Ε, καὶ τὰ ΑΗ, ΓΘ τοῖς Ε, Ζ. διὰ τὰ αὐτὰ δὴ ἴσον ἐστὶ τὸ ΗΒ τῷ Ε, καὶ τὰ ΗΒ, ΘΔ τοῖς Ε, Ζ· ὅσα ἄρα ἐστὶν ἐν τῷ ΑΒ ἴσα τῷ Ε, τοσαῦτα καὶ ἐν τοῖς ΑΒ, ΓΔ ἴσα τοῖς Ε, Ζ· ὁσαπλάσιον ἄρα ἐστὶ τὸ ΑΒ τοῦ Ε, τοσαυταπλάσια ἔσται καὶ τὰ ΑΒ, ΓΔ τῶν Ε, Ζ.

Ἐὰν ἄρα ᾖ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἕκαστον ἑκάστου ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἓν τῶν μεγεθῶν ἑνός, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων· ὅπερ ἔδει δεῖξαι.

times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second).



Let there be any number of magnitudes whatsoever, AB, CD, (which are) equal multiples, respectively, of some (other) magnitudes, E, F, of equal number (to them). I say that as many times as AB is (divisible) by E, so many times will AB, CD also be (divisible) by E, F.

For since AB, CD are equal multiples of E, F, thus as many magnitudes as (there) are in AB equal to E, so many (are there) also in CD equal to F. Let AB have been divided into magnitudes AG, GB, equal to E, and CD into (magnitudes) CH, HD, equal to F. So, the number of (divisions) AG, GB will be equal to the number of (divisions) CH, HD. And since AG is equal to E, and CH to F, AG (is) thus equal to E, and AG, CH to E, F. So, for the same (reasons), GB is equal to E, and GB, HD to E, F. Thus, as many (magnitudes) as (there) are in AB equal to E, so many (are there) also in AB, CD equal to E, F. Thus, as many times as AB is (divisible) by E, so many times will AB, CD also be (divisible) by E, F.

Thus, if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), then as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second). (Which is) the very thing it was required to show.

β′.

Ἐὰν πρῶτον δευτέρου ἰσάκις ᾖ πολλαπλάσιον καὶ τρίτον τετάρτου, ᾖ δὲ καὶ πέμπτον δευτέρου ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτου, καὶ συντεθὲν πρῶτον καὶ πέμπτον δευτέρου ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτου.

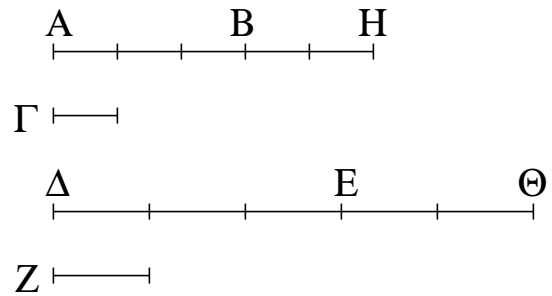
Πρῶτον γὰρ τὸ ΑΒ δευτέρου τοῦ Γ ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ ΔΕ τετάρτου τοῦ Ζ, ἔστω δὲ καὶ πέμπτον τὸ ΒΗ δευτέρου τοῦ Γ ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ ΕΘ τετάρτου τοῦ Ζ· λέγω, ὅτι καὶ συντεθὲν πρῶτον καὶ πέμπτον τὸ ΑΗ δευτέρου τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ ΔΘ τετάρτου τοῦ Ζ.

Proposition 2[†]

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), then the first (magnitude) and the fifth, being added together, and the third and the sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively).

For let a first (magnitude) AB and a third DE be equal multiples of a second C and a fourth F (respectively). And let a fifth (magnitude) BG and a sixth EH also be (other) equal multiples of the second C and the fourth F (respectively). I say that the first (magnitude) and the fifth, being added together, (to give) AG, and the third (magnitude) and the sixth, (being added together,

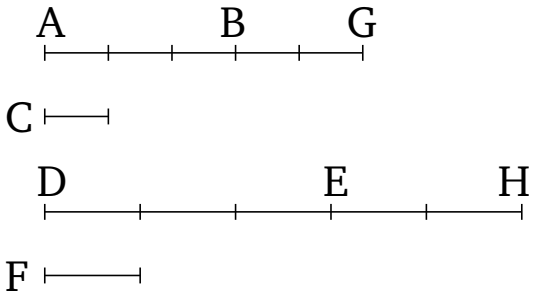
[†] In modern notation, this proposition reads $m\,\alpha + m\,\beta + \cdots = m\,(\alpha + \beta + \cdots)$.



Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Γ καὶ τὸ ΔΕ τοῦ Ζ, ὅσα ἄρα ἐστὶν ἐν τῷ ΑΒ ἴσα τῷ Γ, τοσαῦτα καὶ ἐν τῷ ΔΕ ἴσα τῷ Ζ. διὰ τὰ αὐτὰ δὴ καὶ ὅσα ἐστὶν ἐν τῷ ΒΗ ἴσα τῷ Γ, τοσαῦτα καὶ ἐν τῷ ΕΘ ἴσα τῷ Ζ· ὅσα ἄρα ἐστὶν ἐν ὅλῳ τῷ ΑΗ ἴσα τῷ Γ, τοσαῦτα καὶ ἐν ὅλῳ τῷ ΔΘ ἴσα τῷ Ζ· ὁσαπλάσιον ἄρα ἐστὶ τὸ ΑΗ τοῦ Γ, τοσαυταπλάσιον ἔσται καὶ τὸ ΔΘ τοῦ Ζ. καὶ συντεθὲν ἄρα πρῶτον καὶ πέμπτον τὸ ΑΗ δευτέρου τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ ΔΘ τετάρτου τοῦ Ζ.

Ἐὰν ἄρα πρῶτον δευτέρου ἰσάκις ᾖ πολλαπλάσιον καὶ τρίτον τετάρτου, ᾖ δὲ καὶ πέμπτον δευτέρου ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτου, καὶ συντεθὲν πρῶτον καὶ πέμπτον δευτέρου ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτου· ὅπερ ἔδει δεῖξαι.

to give) DH, will also be equal multiples of the second (magnitude) C and the fourth F (respectively).



For since AB and DE are equal multiples of C and F (respectively), thus as many (magnitudes) as (there) are in AB equal to C, so many (are there) also in DE equal to F. And so, for the same (reasons), as many (magnitudes) as (there) are in BG equal to C, so many (are there) also in EH equal to F. Thus, as many (magnitudes) as (there) are in the whole of AG equal to C, so many (are there) also in the whole of DH equal to F. Thus, as many times as AG is (divisible) by C, so many times will DH also be divisible by F. Thus, the first (magnitude) and the fifth, being added together, (to give) AG, and the third (magnitude) and the sixth, (being added together, to give) DH, will also be equal multiples of the second (magnitude) C and the fourth F (respectively).

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), then the first (magnitude) and the fifth, being added together, and the third and sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively). (Which is) the very thing it was required to show.

† In modern notation, this propostion reads $m\,\alpha + n\,\alpha = (m+n)\,\alpha$.

γ′.

Ἐὰν πρῶτον δευτέρου ἰσάκις ᾖ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου, καὶ δι᾽ ἴσου τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου.

Πρῶτον γὰρ τὸ Α δευτέρου τοῦ Β ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ Γ τετάρτου τοῦ Δ, καὶ εἰλήφθω τῶν Α, Γ ἰσάκις πολλαπλάσια τὰ ΕΖ, ΗΘ· λέγω, ὅτι ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΕΖ τοῦ Β καὶ τὸ ΗΘ τοῦ Δ.

Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΕΖ τοῦ Α καὶ τὸ ΗΘ τοῦ Γ, ὅσα ἄρα ἐστὶν ἐν τῷ ΕΖ ἴσα τῷ Α, τοσαῦτα καὶ ἐν τῷ ΗΘ ἴσα τῷ Γ. διῃρήσθω τὸ μὲν ΕΖ εἰς τὰ τῷ Α μεγέθη ἴσα τὰ ΕΚ, ΚΖ, τὸ δὲ ΗΘ εἰς τὰ τῷ Γ ἴσα τὰ ΗΛ,

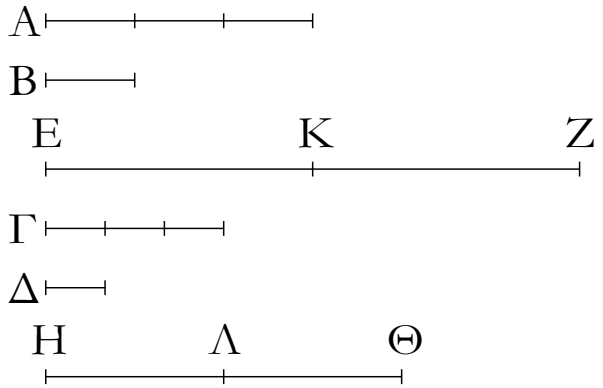
Proposition 3†

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, then, via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively.

For let a first (magnitude) A and a third C be equal multiples of a second B and a fourth D (respectively), and let the equal multiples EF and GH have been taken of A and C (respectively). I say that EF and GH are equal multiples of B and D (respectively).

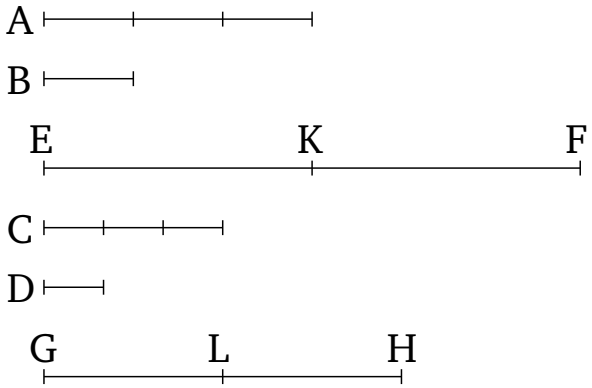
For since EF and GH are equal multiples of A and C (respectively), thus as many (magnitudes) as (there) are in EF equal to A, so many (are there) also in GH

ΛΘ· ἔσται δὴ ἴσον τὸ πλῆθος τῶν ΕΚ, ΚΖ τῷ πλήθει τῶν ΗΛ, ΛΘ. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ Α τοῦ Β καὶ τὸ Γ τοῦ Δ, ἴσον δὲ τὸ μὲν ΕΚ τῷ Α, τὸ δὲ ΗΛ τῷ Γ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΕΚ τοῦ Β καὶ τὸ ΗΛ τοῦ Δ. διὰ τὰ αὐτὰ δὴ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΚΖ τοῦ Β καὶ τὸ ΛΘ τοῦ Δ. ἐπεὶ οὖν πρῶτον τὸ ΕΚ δευτέρου τοῦ Β ἴσάκις ἐστὶ πολλαπλάσιον καὶ τρίτον τὸ ΗΛ τετάρτου τοῦ Δ, ἔστι δὲ καὶ πέμπτον τὸ ΚΖ δευτέρου τοῦ Β ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ ΛΘ τετάρτου τοῦ Δ, καὶ συντεθὲν ἄρα πρῶτον καὶ πέμπτον τὸ ΕΖ δευτέρου τοῦ Β ἰσάκις ἐστὶ πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ ΗΘ τετάρτου τοῦ Δ.



Ἐὰν ἄρα πρῶτον δευτέρου ἰσάκις ᾖ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια, καὶ δι᾽ ἴσου τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου· ὅπερ ἔδει δεῖξαι.

equal to C. Let EF have been divided into magnitudes EK, KF equal to A, and GH into (magnitudes) GL, LH equal to C. So, the number of (magnitudes) EK, KF will be equal to the number of (magnitudes) GL, LH. And since A and C are equal multiples of B and D (respectively), and EK (is) equal to A, and GL to C, EK and GL are thus equal multiples of B and D (respectively). So, for the same (reasons), KF and LH are equal multiples of B and D (respectively). Therefore, since the first (magnitude) EK and the third GL are equal multiples of the second B and the fourth D (respectively), and the fifth (magnitude) KF and the sixth LH are also equal multiples of the second B and the fourth D (respectively), then the first (magnitude) and fifth, being added together, (to give) EF, and the third (magnitude) and sixth, (being added together, to give) GH, are thus also equal multiples of the second (magnitude) B and the fourth D (respectively) [Prop. 5.2].



Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, then, via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m(n\,\alpha) = (m\,n)\,\alpha$.

δ′.

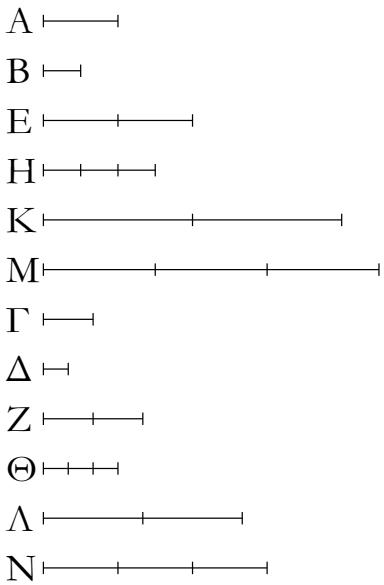
Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου πρὸς τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου καθ᾽ ὁποιονοῦν πολλαπλασιασμὸν τὸν αὐτὸν ἕξει λόγον ληφθέντα κατάλληλα.

Πρῶτον γὰρ τὸ Α πρὸς δεύτερον τὸ Β τὸν αὐτὸν ἐχέτω λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, καὶ εἰλήφθω τῶν μὲν Α, Γ ἰσάκις πολλαπλάσια τὰ Ε, Ζ, τῶν δὲ Β, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Η, Θ· λέγω, ὅτι ἐστὶν ὡς τὸ Ε πρὸς τὸ Η, οὕτως τὸ Ζ πρὸς τὸ Θ.

Proposition 4†

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth then equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D. And let equal multiples E and F have been taken of A and C (respectively), and other random equal multiples G and

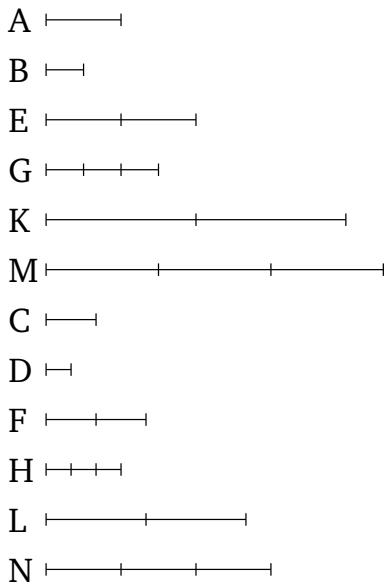


Εἰλήφθω γὰρ τῶν μὲν Ε, Ζ ἰσάκις πολλαπλάσια τὰ Κ, Λ, τῶν δὲ Η, Θ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν.

[Καὶ] ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ μὲν Ε τοῦ Α, τὸ δὲ Ζ τοῦ Γ, καὶ εἴληπται τῶν Ε, Ζ ἰσάκις πολλαπλάσια τὰ Κ, Λ, ἴσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ Κ τοῦ Α καὶ τὸ Λ τοῦ Γ. διὰ τὰ αὐτὰ δὴ ἰσάκις ἐστὶ πολλαπλάσιον τὸ Μ τοῦ Β καὶ τὸ Ν τοῦ Δ. καὶ ἐπεί ἐστιν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ εἴληπται τῶν μὲν Α, Γ ἰσάκις πολλαπλάσια τὰ Κ, Λ, τῶν δὲ Β, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν, εἰ ἄρα ὑπερέχει τὸ Κ τοῦ Μ, ὑπερέχει καὶ τὸ Λ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν Κ, Λ τῶν Ε, Ζ ἰσάκις πολλαπλάσια, τὰ δὲ Μ, Ν τῶν Η, Θ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ Ε πρὸς τὸ Η, οὕτως τὸ Ζ πρὸς τὸ Θ.

Ἐὰν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ τρίτου πρὸς τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου τὸν αὐτὸν ἕξει λόγον καθ᾽ ὁποιονοῦν πολλαπλασιασμὸν ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

H of B and D (respectively). I say that as E (is) to G, so F (is) to H.



For let equal multiples K and L have been taken of E and F (respectively), and other random equal multiples M and N of G and H (respectively).

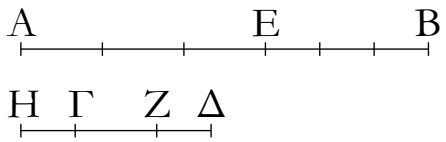
[And] since E and F are equal multiples of A and C (respectively), and the equal multiples K and L have been taken of E and F (respectively), K and L are thus equal multiples of A and C (respectively) [Prop. 5.3]. So, for the same (reasons), M and N are equal multiples of B and D (respectively). And since as A is to B, so C (is) to D, and the equal multiples K and L have been taken of A and C (respectively), and the other random equal multiples M and N of B and D (respectively), then if K exceeds M then L also exceeds N, and if (K is) equal (to M then L is also) equal (to N), and if (K is) less (than M then L is also) less (than N) [Def. 5.5]. And K and L are equal multiples of E and F (respectively), and M and N other random equal multiples of G and H (respectively). Thus, as E (is) to G, so F (is) to H [Def. 5.5].

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth then equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $m\,\alpha : n\,\beta :: m\,\gamma : n\,\delta$, for all m and n.

ε′.

Ἐὰν μέγεθος μεγέθους ἰσάκις ᾖ πολλαπλάσιον, ὅπερ ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστι τὸ ὅλον τοῦ ὅλου.



Μέγεθος γὰρ τὸ ΑΒ μεγέθους τοῦ ΓΔ ἰσάκις ἔστω πολλαπλάσιον, ὅπερ ἀφαιρεθὲν τὸ ΑΕ ἀφαιρεθέντος τοῦ ΓΖ· λέγω, ὅτι καὶ λοιπὸν τὸ ΕΒ λοιποῦ τοῦ ΖΔ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ ΑΒ ὅλου τοῦ ΓΔ.

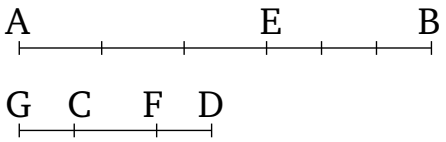
Ὁσαπλάσιον γάρ ἐστι τὸ ΑΕ τοῦ ΓΖ, τοσαυταπλάσιον γεγονέτω καὶ τὸ ΕΒ τοῦ ΓΗ.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΕΒ τοῦ ΗΓ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΑΒ τοῦ ΗΖ. κεῖται δὲ ἰσάκις πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΑΒ τοῦ ΓΔ. ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΑΒ ἑκατέρου τῶν ΗΖ, ΓΔ· ἴσον ἄρα τὸ ΗΖ τῷ ΓΔ. κοινὸν ἀφῃρήσθω τὸ ΓΖ· λοιπὸν ἄρα τὸ ΗΓ λοιπῷ τῷ ΖΔ ἴσον ἐστίν. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΕΒ τοῦ ΗΓ, ἴσον δὲ τὸ ΗΓ τῷ ΔΖ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΕΒ τοῦ ΖΔ. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ ΑΕ τοῦ ΓΖ καὶ τὸ ΑΒ τοῦ ΓΔ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΕΒ τοῦ ΖΔ καὶ τὸ ΑΒ τοῦ ΓΔ. καὶ λοιπὸν ἄρα τὸ ΕΒ λοιποῦ τοῦ ΖΔ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ ΑΒ ὅλου τοῦ ΓΔ.

Ἐὰν ἄρα μέγεθος μεγέθους ἰσάκις ᾖ πολλαπλάσιον, ὅπερ ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστι καὶ τὸ ὅλον τοῦ ὅλου· ὅπερ ἔδει δεῖξαι.

Proposition 5†

If a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively) then the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively).



For let the magnitude AB be the same multiple of the magnitude CD that the (part) taken away AE (is) of the (part) taken away CF (respectively). I say that the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

For as many times as AE is (divisible) by CF, so many times let EB also have been made (divisible) by CG.

And since AE and EB are equal multiples of CF and GC (respectively), AE and AB are thus equal multiples of CF and GF (respectively) [Prop. 5.1]. And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, AB is an equal multiple of each of GF and CD. Thus, GF (is) equal to CD. Let CF have been subtracted from both. Thus, the remainder GC is equal to the remainder FD. And since AE and EB are equal multiples of CF and GC (respectively), and GC (is) equal to DF, AE and EB are thus equal multiples of CF and FD (respectively). And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, EB and AB are equal multiples of FD and CD (respectively). Thus, the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

Thus, if a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively) then the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\,\alpha - m\,\beta = m\,(\alpha - \beta)$.

ϛ′.

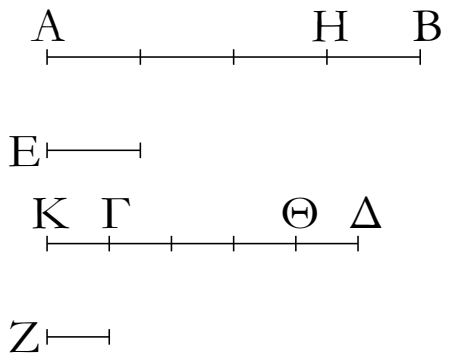
Ἐὰν δύο μεγέθη δύο μεγεθῶν ἰσάκις ᾖ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ᾖ πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς ἤτοι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Δύο γὰρ μεγέθη τὰ ΑΒ, ΓΔ δύο μεγεθῶν τῶν Ε, Ζ

Proposition 6†

If two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), then the remainders are also either equal to the latter (magnitudes), or (are) equal multiples

ἰσάκις ἔστω πολλαπλάσια, καὶ ἀφαιρεθέντα τὰ ΑΗ, ΓΘ τῶν αὐτῶν τῶν Ε, Ζ ἰσάκις ἔστω πολλαπλάσια· λέγω, ὅτι καὶ λοιπὰ τὰ ΗΒ, ΘΔ τοῖς Ε, Ζ ἤτοι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.



Ἔστω γὰρ πρότερον τὸ ΗΒ τῷ Ε ἴσον· λέγω, ὅτι καὶ τὸ ΘΔ τῷ Ζ ἴσον ἐστίν.

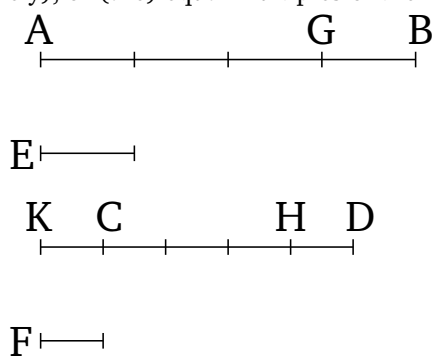
Κείσθω γὰρ τῷ Ζ ἴσον τὸ ΓΚ. ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΗ τοῦ Ε καὶ τὸ ΓΘ τοῦ Ζ, ἴσον δὲ τὸ μὲν ΗΒ τῷ Ε, τὸ δὲ ΚΓ τῷ Ζ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Ε καὶ τὸ ΚΘ τοῦ Ζ. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ ΑΒ τοῦ Ε καὶ τὸ ΓΔ τοῦ Ζ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΚΘ τοῦ Ζ καὶ τὸ ΓΔ τοῦ Ζ. ἐπεὶ οὖν ἑκάτερον τῶν ΚΘ, ΓΔ τοῦ Ζ ἰσάκις ἐστὶ πολλαπλάσιον, ἴσον ἄρα ἐστὶ τὸ ΚΘ τῷ ΓΔ. κοινὸν ἀφῃρήσθω τὸ ΓΘ· λοιπὸν ἄρα τὸ ΚΓ λοιπῷ τῷ ΘΔ ἴσον ἐστίν. ἀλλὰ τὸ Ζ τῷ ΚΓ ἐστιν ἴσον· καὶ τὸ ΘΔ ἄρα τῷ Ζ ἴσον ἐστίν. ὥστε εἰ τὸ ΗΒ τῷ Ε ἴσον ἐστίν, καὶ τὸ ΘΔ ἴσον ἔσται τῷ Ζ.

Ὁμοίως δὴ δείξομεν, ὅτι, κἂν πολλαπλάσιον ᾖ τὸ ΗΒ τοῦ Ε, τοσαυταπλάσιον ἔσται καὶ τὸ ΘΔ τοῦ Ζ.

Ἐὰν ἄρα δύο μεγέθη δύο μεγεθῶν ἰσάκις ᾖ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ᾖ πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς ἤτοι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια· ὅπερ ἔδει δεῖξαι.

of them (respectively).

For let two magnitudes AB and CD be equal multiples of two magnitudes E and F (respectively). And let the (parts) taken away (from the former) AG and CH be equal multiples of E and F (respectively). I say that the remainders GB and HD are also either equal to E and F (respectively), or (are) equal multiples of them.



For let GB be, first of all, equal to E. I say that HD is also equal to F.

For let CK be made equal to F. Since AG and CH are equal multiples of E and F (respectively), and GB (is) equal to E, and KC to F, AB and KH are thus equal multiples of E and F (respectively) [Prop. 5.2]. And AB and CD are assumed (to be) equal multiples of E and F (respectively). Thus, KH and CD are equal multiples of F and F (respectively). Therefore, KH and CD are each equal multiples of F. Thus, KH is equal to CD. Let CH have be taken away from both. Thus, the remainder KC is equal to the remainder HD. But, F is equal to KC. Thus, HD is also equal to F. Hence, if GB is equal to E then HD will also be equal to F.

So, similarly, we can show that even if GB is a multiple of E then HD will also be the same multiple of F.

Thus, if two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), then the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\,\alpha - n\,\alpha = (m-n)\,\alpha$.

ζ′.

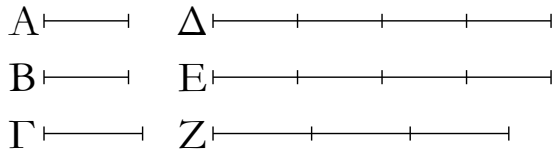
Τὰ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

Ἔστω ἴσα μεγέθη τὰ Α, Β, ἄλλο δέ τι, ὃ ἔτυχεν, μέγεθος τὸ Γ· λέγω, ὅτι ἑκάτερον τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν ἔχει λόγον, καὶ τὸ Γ πρὸς ἑκάτερον τῶν Α, Β.

Proposition 7

Equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Let A and B be equal magnitudes, and C some other random magnitude. I say that A and B each have the



Εἰλήφθω γὰρ τῶν μὲν Α, Β ἰσάκις πολλαπλάσια τὰ Δ, Ε, τοῦ δὲ Γ ἄλλο, ὃ ἔτυχεν, πολλαπλάσιον τὸ Ζ.

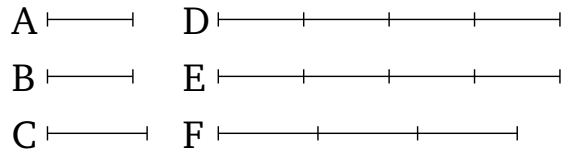
Ἐπεὶ οὖν ἰσάκις ἐστὶ πολλαπλάσιον τὸ Δ τοῦ Α καὶ τὸ Ε τοῦ Β, ἴσον δὲ τὸ Α τῷ Β, ἴσον ἄρα καὶ τὸ Δ τῷ Ε. ἄλλο δέ, ὃ ἔτυχεν, τὸ Ζ. Εἰ ἄρα ὑπερέχει τὸ Δ τοῦ Ζ, ὑπερέχει καὶ τὸ Ε τοῦ Ζ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν Δ, Ε τῶν Α, Β ἰσάκις πολλαπλάσια, τὸ δὲ Ζ τοῦ Γ ἄλλο, ὃ ἔτυχεν, πολλαπλάσιον· ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Γ, οὕτως τὸ Β πρὸς τὸ Γ.

Λέγω [δή], ὅτι καὶ τὸ Γ πρὸς ἑκάτερον τῶν Α, Β τὸν αὐτὸν ἔχει λόγον.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείξομεν, ὅτι ἴσον ἐστὶ τὸ Δ τῷ Ε· ἄλλο δέ τι τὸ Ζ· εἰ ἄρα ὑπερέχει τὸ Ζ τοῦ Δ, ὑπερέχει καὶ τοῦ Ε, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὸ μὲν Ζ τοῦ Γ πολλαπλάσιον, τὰ δὲ Δ, Ε τῶν Α, Β ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Α, οὕτως τὸ Γ πρὸς τὸ Β.

Τὰ ἴσα ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

same ratio to C, and (that) C (has the same ratio) to each of A and B.



For let the equal multiples D and E have been taken of A and B (respectively), and the other random multiple F of C.

Therefore, since D and E are equal multiples of A and B (respectively), and A (is) equal to B, D (is) thus also equal to E. And F (is) different, at random. Thus, if D exceeds F then E also exceeds F, and if (D is) equal (to F then E is also) equal (to F), and if (D is) less (than F then E is also) less (than F). And D and E are equal multiples of A and B (respectively), and F another random multiple of C. Thus, as A (is) to C, so B (is) to C [Def. 5.5].

[So] I say that C† also has the same ratio to each of A and B.

For, similarly, we can show, by the same construction, that D is equal to E. And F (has) some other (value). Thus, if F exceeds D then it also exceeds E, and if (F is) equal (to D then it is also) equal (to E), and if (F is) less (than D then it is also) less (than E). And F is a multiple of C, and D and E other random equal multiples of A and B. Thus, as C (is) to A, so C (is) to B [Def. 5.5].

Thus, equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐὰν μεγέθη τινὰ ἀνάλογον ᾖ, καὶ ἀνάπαλιν ἀνάλογον ἔσται. ὅπερ ἔδει δεῖξαι.

Corollary‡

So (it is) clear, from this, that if some magnitudes are proportional then they will also be proportional inversely. (Which is) the very thing it was required to show.

† The Greek text has "E", which is obviously a mistake.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\beta : \alpha :: \delta : \gamma$.

η′.

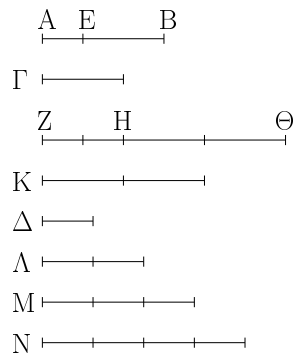
Τῶν ἀνίσων μεγεθῶν τὸ μεῖζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔχει ἤπερ τὸ ἔλαττον. καὶ τὸ αὐτὸ πρὸς τὸ ἔλαττον μείζονα λόγον ἔχει ἤπερ πρὸς τὸ μεῖζον.

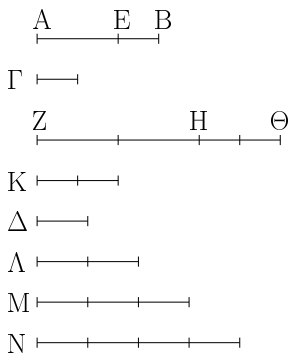
Ἔστω ἄνισα μεγέθη τὰ ΑΒ, Γ, καὶ ἔστω μεῖζον τὸ ΑΒ, ἄλλο δέ, ὃ ἔτυχεν, τὸ Δ· λέγω, ὅτι τὸ ΑΒ πρὸς τὸ Δ μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Δ, καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔχει ἤπερ πρὸς τὸ ΑΒ.

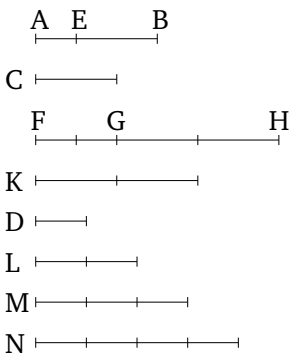
Proposition 8

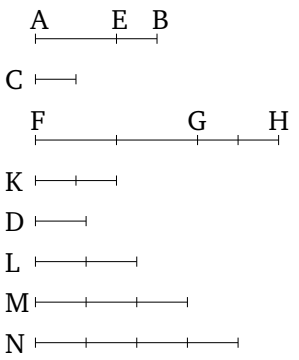
For unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater.

Let AB and C be unequal magnitudes, and let AB be the greater (of the two), and D another random magnitude. I say that AB has a greater ratio to D than C (has) to D, and (that) D has a greater ratio to C than (it has) to AB.









Ἐπεὶ γὰρ μεῖζόν ἐστι τὸ ΑΒ τοῦ Γ, κείσθω τῷ Γ ἴσον τὸ ΒΕ· τὸ δὴ ἔλασσον τῶν ΑΕ, ΕΒ πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ Δ μεῖζον. ἔστω πρότερον τὸ ΑΕ ἔλαττον τοῦ ΕΒ, καὶ πεπολλαπλασιάσθω τὸ ΑΕ, καὶ ἔστω αὐτοῦ πολλαπλάσιον τὸ ΖΗ μεῖζον ὂν τοῦ Δ, καὶ ὁσαπλάσιόν ἐστι τὸ ΖΗ τοῦ ΑΕ, τοσαυταπλάσιον γεγονέτω καὶ τὸ μὲν ΗΘ τοῦ ΕΒ τὸ δὲ Κ τοῦ Γ· καὶ εἰλήφθω τοῦ Δ διπλάσιον μὲν τὸ Λ, τριπλάσιον δὲ τὸ Μ, καὶ ἑξῆς ἑνὶ πλεῖον, ἕως ἂν τὸ λαμβανόμενον πολλαπλάσιον μὲν γένηται τοῦ Δ, πρώτως δὲ μεῖζον τοῦ Κ. εἰλήφθω, καὶ ἔστω τὸ Ν τετραπλάσιον μὲν τοῦ Δ, πρώτως δὲ μεῖζον τοῦ Κ.

Ἐπεὶ οὖν τὸ Κ τοῦ Ν πρώτως ἐστὶν ἔλαττον, τὸ Κ ἄρα τοῦ Μ οὐκ ἐστιν ἔλαττον. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΖΗ τοῦ ΑΕ καὶ τὸ ΗΘ τοῦ ΕΒ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΖΗ τοῦ ΑΕ καὶ τὸ ΖΘ τοῦ ΑΒ. ἰσάκις δέ ἐστι πολλαπλάσιον τὸ ΖΗ τοῦ ΑΕ καὶ τὸ Κ τοῦ Γ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΖΘ τοῦ ΑΒ καὶ τὸ Κ τοῦ Γ. τὰ ΖΘ, Κ ἄρα τῶν ΑΒ, Γ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΕΒ καὶ τὸ Κ τοῦ Γ, ἴσον δὲ τὸ ΕΒ τῷ Γ, ἴσον ἄρα καὶ τὸ ΗΘ τῷ Κ. τὸ δὲ Κ τοῦ Μ οὐκ ἐστιν ἔλαττον· οὐδ᾽ ἄρα τὸ ΗΘ τοῦ Μ ἔλαττόν ἐστιν. μεῖζον δὲ τὸ ΖΗ τοῦ Δ· ὅλον ἄρα τὸ ΖΘ συναμφοτέρων τῶν Δ, Μ μεῖζόν ἐστιν. ἀλλὰ συναμφότερα τὰ Δ, Μ τῷ Ν ἐστιν ἴσα, ἐπειδήπερ τὸ Μ τοῦ Δ τριπλάσιόν ἐστιν, συναμφότερα δὲ τὰ Μ, Δ τοῦ Δ ἐστι τετραπλάσια, ἔστι δὲ καὶ τὸ Ν τοῦ Δ τετραπλάσιον· συναμφότερα ἄρα τὰ Μ, Δ τῷ Ν ἴσα ἐστίν. ἀλλὰ τὸ ΖΘ τῶν Μ, Δ μεῖζόν ἐστιν· τὸ ΖΘ ἄρα τοῦ Ν ὑπερέχει· τὸ δὲ Κ τοῦ Ν οὐχ ὑπερέχει. καί ἐστι τὰ μὲν ΖΘ, Κ τῶν ΑΒ, Γ ἰσάκις πολλαπλάσια, τὸ δὲ Ν τοῦ Δ ἄλλο, ὃ ἔτυχεν, πολλαπλάσιον· τὸ ΑΒ ἄρα πρὸς τὸ Δ μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Δ.

Λέγω δή, ὅτι καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔχει ἤπερ τὸ Δ πρὸς τὸ ΑΒ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείξομεν, ὅτι τὸ μὲν Ν τοῦ Κ ὑπερέχει, τὸ δὲ Ν τοῦ ΖΘ οὐχ ὑπερέχει. καί ἐστι τὸ μὲν Ν τοῦ Δ πολλαπλάσιον, τὰ δὲ ΖΘ, Κ τῶν ΑΒ, Γ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· τὸ Δ ἄρα πρὸς τὸ Γ μείζονα λόγον ἔχει ἤπερ τὸ Δ πρὸς τὸ ΑΒ.

Ἀλλὰ δὴ τὸ ΑΕ τοῦ ΕΒ μεῖζον ἔστω. τὸ δὴ ἔλαττον τὸ ΕΒ πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ Δ μεῖζον. πε-

For since AB is greater than C, let BE be made equal to C. So, the lesser of AE and EB, being multiplied, will sometimes be greater than D [Def. 5.4]. First of all, let AE be less than EB, and let AE have been multiplied, and let FG be a multiple of it which (is) greater than D. And as many times as FG is (divisible) by AE, so many times let GH also have become (divisible) by EB, and K by C. And let the double multiple L of D have been taken, and the triple multiple M, and several more, (each increasing) in order by one, until the (multiple) taken becomes the first multiple of D (which is) greater than K. Let it have been taken, and let it also be the quadruple multiple N of D—the first (multiple) greater than K.

Therefore, since K is less than N first, K is thus not less than M. And since FG and GH are equal multiples of AE and EB (respectively), FG and FH are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. And FG and K are equal multiples of AE and C (respectively). Thus, FH and K are equal multiples of AB and C (respectively). Thus, FH, K are equal multiples of AB, C. Again, since GH and K are equal multiples of EB and C, and EB (is) equal to C, GH (is) thus also equal to K. And K is not less than M. Thus, GH not less than M either. And FG (is) greater than D. Thus, the whole of FH is greater than D and M (added) together. But, D and M (added) together is equal to N, inasmuch as M is three times D, and M and D (added) together is four times D, and N is also four times D. Thus, M and D (added) together is equal to N. But, FH is greater than M and D. Thus, FH exceeds N. And K does not exceed N. And FH, K are equal multiples of AB, C, and N another random multiple of D. Thus, AB has a greater ratio to D than C (has) to D [Def. 5.7].

So, I say that D also has a greater ratio to C than D (has) to AB.

For, similarly, by the same construction, we can show that N exceeds K, and N does not exceed FH. And N is a multiple of D, and FH, K other random equal multiples of AB, C (respectively). Thus, D has a greater

πολλαπλασιάσθω, καὶ ἔστω τὸ ΗΘ πολλαπλάσιον μὲν τοῦ ΕΒ, μεῖζον δὲ τοῦ Δ· καὶ ὁσαπλάσιόν ἐστι τὸ ΗΘ τοῦ ΕΒ, τοσαυταπλάσιον γεγονέτω καὶ τὸ μὲν ΖΗ τοῦ ΑΕ, τὸ δὲ Κ τοῦ Γ. ὁμοίως δὴ δείξομεν, ὅτι τὰ ΖΘ, Κ τῶν ΑΒ, Γ ἰσάκις ἐστὶ πολλαπλάσια· καὶ εἰλήφθω ὁμοίως τὸ Ν πολλαπλάσιον μὲν τοῦ Δ, πρώτως δὲ μεῖζον τοῦ ΖΗ· ὥστε πάλιν τὸ ΖΗ τοῦ Μ οὔκ ἐστιν ἔλασσον. μεῖζον δὲ τὸ ΗΘ τοῦ Δ· ὅλον ἄρα τὸ ΖΘ τῶν Δ, Μ, τουτέστι τοῦ Ν, ὑπερέχει. τὸ δὲ Κ τοῦ Ν οὐχ ὑπερέχει, ἐπειδήπερ καὶ τὸ ΖΗ μεῖζον ὂν τοῦ ΗΘ, τουτέστι τοῦ Κ, τοῦ Ν οὐχ ὑπερέχει. καὶ ὡσαύτως κατακολουθοῦντες τοῖς ἐπάνω περαίνομεν τὴν ἀπόδειξιν.

Τῶν ἄρα ἀνίσων μεγεθῶν τὸ μεῖζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔχει ἤπερ τὸ ἔλαττον· καὶ τὸ αὐτὸ πρὸς τὸ ἔλαττον μείζονα λόγον ἔχει ἤπερ πρὸς τὸ μεῖζον· ὅπερ ἔδει δεῖξαι.

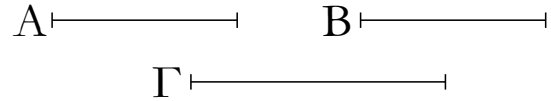
ratio to C than D (has) to AB [Def. 5.5].

And so let AE be greater than EB. So, the lesser, EB, being multiplied, will sometimes be greater than D. Let it have been multiplied, and let GH be a multiple of EB (which is) greater than D. And as many times as GH is (divisible) by EB, so many times let FG also have become (divisible) by AE, and K by C. So, similarly (to the above), we can show that FH and K are equal multiples of AB and C (respectively). And, similarly (to the above), let the multiple N of D, (which is) the first (multiple) greater than FG, have been taken. So, FG is again not less than M. And GH (is) greater than D. Thus, the whole of FH exceeds D and M, that is to say N. And K does not exceed N, inasmuch as FG, which (is) greater than GH—that is to say, K—also does not exceed N. And, following the above (arguments), we (can) complete the proof in the same manner.

Thus, for unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater. (Which is) the very thing it was required to show.

ϑ′.

Τὰ πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον, ἐκεῖνα ἴσα ἐστίν.



Ἐχέτω γὰρ ἑκάτερον τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ Α τῷ Β.

Εἰ γὰρ μή, οὐκ ἂν ἑκάτερον τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν εἶχε λόγον· ἔχει δέ· ἴσον ἄρα ἐστὶ τὸ Α τῷ Β.

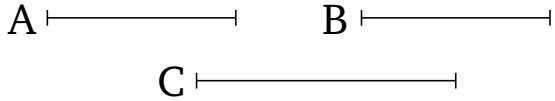
Ἐχέτω δὴ πάλιν τὸ Γ πρὸς ἑκάτερον τῶν Α, Β τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ Α τῷ Β.

Εἰ γὰρ μή, οὐκ ἂν τὸ Γ πρὸς ἑκάτερον τῶν Α, Β τὸν αὐτὸν εἶχε λόγον· ἔχει δέ· ἴσον ἄρα ἐστὶ τὸ Α τῷ Β.

Τὰ ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον, ἐκεῖνα ἴσα ἐστίν· ὅπερ ἔδει δεῖξαι.

Proposition 9

(Magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal.



For let A and B each have the same ratio to C. I say that A is equal to B.

For if not, A and B would not each have the same ratio to C [Prop. 5.8]. But they do. Thus, A is equal to B.

So, again, let C have the same ratio to each of A and B. I say that A is equal to B.

For if not, C would not have the same ratio to each of A and B [Prop. 5.8]. But it does. Thus, A is equal to B.

Thus, (magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal. (Which is) the very thing it was required to show.

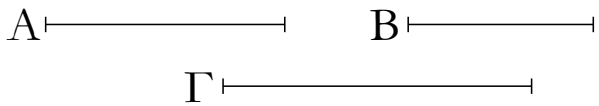
ι′.

Τῶν πρὸς τὸ αὐτὸ λόγον ἐχόντων τὸ μείζονα λόγον ἔχον ἐκεῖνο μεῖζόν ἐστιν· πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον

Proposition 10

For (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is

ἔχει, ἐκεῖνο ἔλαττόν ἐστιν.



Ἐχέτω γὰρ τὸ Α πρὸς τὸ Γ μείζονα λόγον ἤπερ τὸ Β πρὸς τὸ Γ· λέγω, ὅτι μεῖζόν ἐστι τὸ Α τοῦ Β.

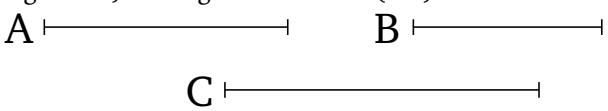
Εἰ γὰρ μή, ἤτοι ἴσον ἐστὶ τὸ Α τῷ Β ἢ ἔλασσον. ἴσον μὲν οὖν οὐκ ἐστὶ τὸ Α τῷ Β· ἑκάτερον γὰρ ἂν τῶν Α, Β πρὸς τὸ Γ τὸν αὐτὸν εἶχε λόγον. οὐκ ἔχει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ Α τῷ Β. οὐδὲ μὴν ἔλασσόν ἐστι τὸ Α τοῦ Β· τὸ Α γὰρ ἂν πρὸς τὸ Γ ἐλάσσονα λόγον εἶχεν ἤπερ τὸ Β πρὸς τὸ Γ. οὐκ ἔχει δέ· οὐκ ἄρα ἔλασσόν ἐστι τὸ Α τοῦ Β. ἐδείχθη δὲ οὐδὲ ἴσον· μεῖζον ἄρα ἐστὶ τὸ Α τοῦ Β.

Ἐχέτω δὴ πάλιν τὸ Γ πρὸς τὸ Β μείζονα λόγον ἤπερ τὸ Γ πρὸς τὸ Α· λέγω, ὅτι ἔλασσόν ἐστι τὸ Β τοῦ Α.

Εἰ γὰρ μή, ἤτοι ἴσον ἐστὶν ἢ μεῖζον. ἴσον μὲν οὖν οὐκ ἐστι τὸ Β τῷ Α· τὸ Γ γὰρ ἂν πρὸς ἑκάτερον τῶν Α, Β τὸν αὐτὸν εἶχε λόγον. οὐκ ἔχει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ Α τῷ Β. οὐδὲ μὴν μεῖζόν ἐστι τὸ Β τοῦ Α· τὸ Γ γὰρ ἂν πρὸς τὸ Β ἐλάσσονα λόγον εἶχεν ἤπερ πρὸς τὸ Α. οὐκ ἔχει δέ· οὐκ ἄρα μεῖζόν ἐστι τὸ Β τοῦ Α. ἐδείχθη δέ, ὅτι οὐδὲ ἴσον· ἔλαττον ἄρα ἐστὶ τὸ Β τοῦ Α.

Τῶν ἄρα πρὸς τὸ αὐτὸ λόγον ἐχόντων τὸ μείζονα λόγον ἔχον μεῖζόν ἐστιν· καὶ πρὸς ὃ τὸ αὐτὸ μείζονα λόγον ἔχει, ἐκεῖνο ἔλαττόν ἐστιν· ὅπερ ἔδει δεῖξαι.

(the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser.



For let A have a greater ratio to C than B (has) to C. I say that A is greater than B.

For if not, A is surely either equal to or less than B. In fact, A is not equal to B. For (then) A and B would each have the same ratio to C [Prop. 5.7]. But they do not. Thus, A is not equal to B. Neither, indeed, is A less than B. For (then) A would have a lesser ratio to C than B (has) to C [Prop. 5.8]. But it does not. Thus, A is not less than B. And it was shown not (to be) equal either. Thus, A is greater than B.

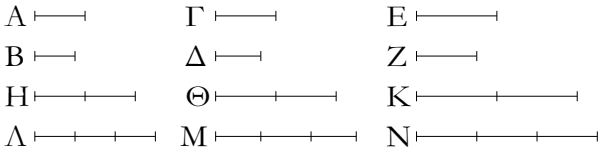
So, again, let C have a greater ratio to B than C (has) to A. I say that B is less than A.

For if not, (it is) surely either equal or greater. In fact, B is not equal to A. For (then) C would have the same ratio to each of A and B [Prop. 5.7]. But it does not. Thus, A is not equal to B. Neither, indeed, is B greater than A. For (then) C would have a lesser ratio to B than (it has) to A [Prop. 5.8]. But it does not. Thus, B is not greater than A. And it was shown that (it is) not equal (to A) either. Thus, B is less than A.

Thus, for (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser. (Which is) the very thing it was required to show.

ια′.

Οἱ τῷ αὐτῷ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί.



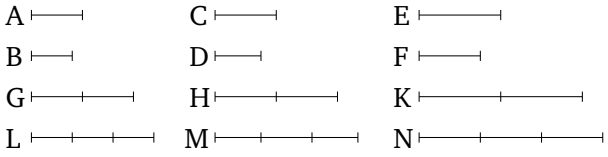
Ἔστωσαν γὰρ ὡς μὲν τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, ὡς δὲ τὸ Γ πρὸς τὸ Δ, οὕτως τὸ Ε πρὸς τὸ Ζ· λέγω, ὅτι ἐστὶν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Ε πρὸς τὸ Ζ.

Εἰλήφθω γὰρ τῶν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ, τῶν δὲ Β, Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν.

Καὶ ἐπεί ἐστιν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ εἴληπται τῶν μὲν Α, Γ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, εἰ ἄρα ὑπερέχει τὸ Η τοῦ Λ, ὑπερέχει καὶ τὸ Θ τοῦ Μ, καὶ εἰ ἴσον ἐστίν, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. πάλιν, ἐπεί ἐστιν

Proposition 11[†]

(Ratios which are) the same with the same ratio are also the same with one another.



For let it be that as A (is) to B, so C (is) to D, and as C (is) to D, so E (is) to F. I say that as A is to B, so E (is) to F.

For let the equal multiples G, H, K have been taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B, so C (is) to D, and the equal multiples G and H have been taken of A and C (respectively), and the other random equal multiples L and M of B and D (respectively), thus if G exceeds L then H also exceeds M, and if (G is) equal (to L then H is also)

ὡς τὸ Γ πρὸς τὸ Δ, οὕτως τὸ Ε πρὸς τὸ Ζ, καὶ εἵληπται τῶν Γ, Ε ἰσάκις πολλαπλάσια τὰ Θ, Κ, τῶν δὲ Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν, εἰ ἄρα ὑπερέχει τὸ Θ τοῦ Μ, ὑπερέχει καὶ τὸ Κ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλλατον, ἔλαττον. ἀλλὰ εἰ ὑπερεῖχε τὸ Θ τοῦ Μ, ὑπερεῖχε καὶ τὸ Η τοῦ Λ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον· ὥστε καὶ εἰ ὑπερέχει τὸ Η τοῦ Λ, ὑπερέχει καὶ τὸ Κ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν Η, Κ τῶν Α, Ε ἰσάκις πολλαπλάσια, τὰ δὲ Λ, Ν τῶν Β, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Ε πρὸς τὸ Ζ.

Οἱ ἄρα τῷ αὐτῷ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί· ὅπερ ἔδει δεῖξαι.

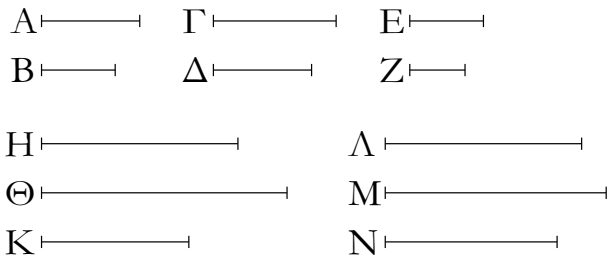
equal (to M), and if (G is) less (than L then H is also) less (than M) [Def. 5.5]. Again, since as C is to D, so E (is) to F, and the equal multiples H and K have been taken of C and E (respectively), and the other random equal multiples M and N of D and F (respectively), thus if H exceeds M then K also exceeds N, and if (H is) equal (to M then K is also) equal (to N), and if (H is) less (than M then K is also) less (than N) [Def. 5.5]. But (we saw that) if H was exceeding M then G was also exceeding L, and if (H was) equal (to M then G was also) equal (to L), and if (H was) less (than M then G was also) less (than L). And, hence, if G exceeds L then K also exceeds N, and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N). And G and K are equal multiples of A and E (respectively), and L and N other random equal multiples of B and F (respectively). Thus, as A is to B, so E (is) to F [Def. 5.5].

Thus, (ratios which are) the same with the same ratio are also the same with one another. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta :: \epsilon : \zeta$ then $\alpha : \beta :: \epsilon : \zeta$.

ιβ′.

Ἐὰν ᾖ ὁποσαοῦν μεγέθη ἀνάλογον, ἔσται ὡς ἓν τῶν ἡγουμένων πρὸς ἓν τῶν ἑπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἑπόμενα.



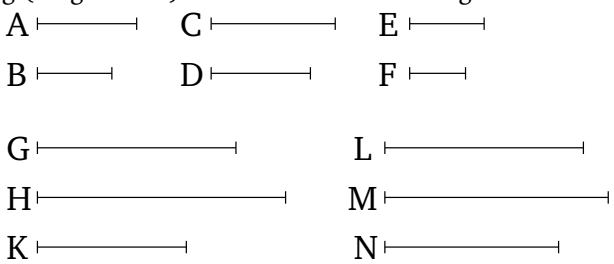
Ἔστωσαν ὁποσαοῦν μεγέθη ἀνάλογον τὰ Α, Β, Γ, Δ, Ε, Ζ, ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς το Ζ· λέγω, ὅτι ἐστὶν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Εἰλήφθω γὰρ τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ, τῶν δὲ Β, Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν.

Καὶ ἐπεί ἐστιν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς τὸ Ζ, καὶ εἵληπται τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ τῶν δὲ Β, Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν, εἰ ἄρα ὑπερέχει τὸ Η τοῦ Λ, ὑπερέχει καὶ τὸ Θ τοῦ Μ, καὶ τὸ Κ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὥστε καὶ εἰ ὑπερέχει τὸ Η τοῦ Λ,

Proposition 12†

If there are any number of magnitudes whatsoever (which are) proportional then as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following.



Let there be any number of magnitudes whatsoever, A, B, C, D, E, F, (which are) proportional, (so that) as A (is) to B, so C (is) to D, and E to F. I say that as A is to B, so A, C, E (are) to B, D, F.

For let the equal multiples G, H, K have been taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B, so C (is) to D, and E to F, and the equal multiples G, H, K have been taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively), thus if G exceeds L then H also exceeds M, and K (exceeds) N, and if (G is) equal (to L then H is also) equal (to M, and K to N),

ὑπερέχει καὶ τὰ Η, Θ, Κ τῶν Λ, Μ, Ν, καὶ εἰ ἴσον, ἴσα, καὶ εἰ ἔλαττον, ἔλαττονα. καί ἐστι τὸ μὲν Η καὶ τὰ Η, Θ, Κ τοῦ Α καὶ τῶν Α, Γ, Ε ἰσάκις πολλαπλάσια, ἐπειδήπερ ἐὰν ᾖ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἕκαστον ἑκάστου ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἓν τῶν μεγεθῶν ἑνός, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων. διὰ τὰ αὐτὰ δὴ καὶ τὸ Λ καὶ τὰ Λ, Μ, Ν τοῦ Β καὶ τῶν Β, Δ, Ζ ἰσάκις ἐστὶ πολλαπλάσια· ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Β, οὕτως τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Ἐὰν ἄρα ᾖ ὁποσαοῦν μεγέθη ἀνάλογον, ἔσται ὡς ἓν τῶν ἡγουμένων πρὸς ἓν τῶν ἑπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἑπόμενα· ὅπερ ἔδει δεῖξαι.

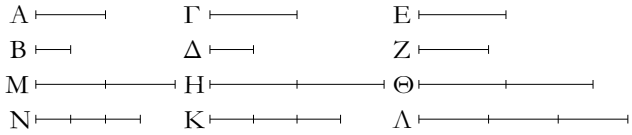
and if (G is) less (than L then H is also) less (than M, and K than N) [Def. 5.5]. And, hence, if G exceeds L then G, H, K also exceed L, M, N, and if (G is) equal (to L then G, H, K are also) equal (to L, M, N) and if (G is) less (than L then G, H, K are also) less (than L, M, N). And G and G, H, K are equal multiples of A and A, C, E (respectively), inasmuch as if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), then as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second) [Prop. 5.1]. So, for the same (reasons), L and L, M, N are also equal multiples of B and B, D, F (respectively). Thus, as A is to B, so A, C, E (are) to B, D, F (respectively).

Thus, if there are any number of magnitudes whatsoever (which are) proportional then as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \alpha' :: \beta : \beta' :: \gamma : \gamma'$ *etc.* then $\alpha : \alpha' :: (\alpha + \beta + \gamma + \cdots) : (\alpha' + \beta' + \gamma' + \cdots)$.

ιγ′.

Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔχῃ ἤ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἕξει ἤ πέμπτον πρὸς ἕκτον.

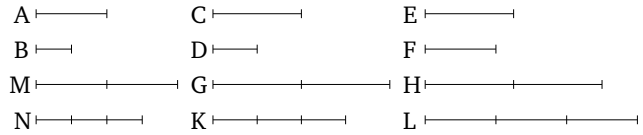


Πρῶτον γὰρ τὸ Α πρὸς δεύτερον τὸ Β τὸν αὐτὸν ἐχέτω λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, τρίτον δὲ τὸ Γ πρὸς τέταρτον τὸ Δ μείζονα λόγον ἐχέτω ἤ πέμπτον τὸ Ε πρὸς ἕκτον τὸ Ζ. λέγω, ὅτι καὶ πρῶτον τὸ Α πρὸς δεύτερον τὸ Β μείζονα λόγον ἕξει ἤπερ πέμπτον τὸ Ε πρὸς ἕκτον τὸ Ζ.

Ἐπεὶ γὰρ ἔστι τινὰ τῶν μὲν Γ, Ε ἰσάκις πολλαπλάσια, τῶν δὲ Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια, καὶ τὸ μὲν τοῦ Γ πολλαπλάσιον τοῦ τοῦ Δ πολλαπλασίου ὑπερέχει, τὸ δὲ τοῦ Ε πολλαπλάσιον τοῦ τοῦ Ζ πολλαπλασίου οὐχ ὑπερέχει, εἰλήφθω, καὶ ἔστω τῶν μὲν Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Δ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Κ, Λ, ὥστε τὸ μὲν Η τοῦ Κ ὑπερέχειν, τὸ δὲ Θ τοῦ Λ μὴ ὑπερέχειν· καὶ ὁσαπλάσιον μέν ἐστι τὸ Η τοῦ Γ, τοσαυταπλάσιον ἔστω καὶ τὸ Μ τοῦ Α, ὁσαπλάσιον δὲ τὸ Κ τοῦ Δ, τοσαυταπλάσιον ἔστω καὶ τὸ Ν τοῦ Β.

Proposition 13†

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the third (magnitude) has a greater ratio to the fourth than a fifth (has) to a sixth, then the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth.



For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D, and let the third (magnitude) C have a greater ratio to the fourth D than a fifth E (has) to a sixth F. I say that the first (magnitude) A will also have a greater ratio to the second B than the fifth E (has) to the sixth F.

For since there are some equal multiples of C and E, and other random equal multiples of D and F, (for which) the multiple of C exceeds the (multiple) of D, and the multiple of E does not exceed the multiple of F [Def. 5.7], let them have been taken. And let G and H be equal multiples of C and E (respectively), and K and L other random equal multiples of D and F (respectively), such that G exceeds K, but H does not exceed L. And as many times as G is (divisible) by C, so many times let M be (divisible) by A. And as many times as K (is divisible)

Καὶ ἐπεί ἐστιν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ εἴληπται τῶν μὲν Α, Γ ἰσάκις πολλαπλάσια τὰ Μ, Η, τῶν δὲ Β, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Ν, Κ, εἰ ἄρα ὑπερέχει τὸ Μ τοῦ Ν, ὑπερέχει καὶ τὸ Η τοῦ Κ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλλατον. ὑπερέχει δὲ τὸ Η τοῦ Κ· ὑπερέχει ἄρα καὶ τὸ Μ τοῦ Ν. τὸ δὲ Θ τοῦ Λ οὐχ ὑπερέχει· καί ἐστι τὰ μὲν Μ, Θ τῶν Α, Ε ἰσάκις πολλαπλάσια, τὰ δὲ Ν, Λ τῶν Β, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· τὸ ἄρα Α πρὸς τὸ Β μείζονα λόγον ἔχει ἤπερ τὸ Ε πρὸς τὸ Ζ.

Ἐὰν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔχῃ ἤ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἕξει ἤ πέμπτον πρὸς ἕκτον· ὅπερ ἔδει δεῖξαι.

by D, so many times let N be (divisible) by B.

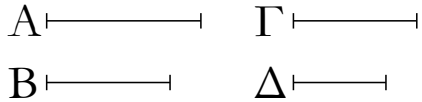
And since as A is to B, so C (is) to D, and the equal multiples M and G have been taken of A and C (respectively), and the other random equal multiples N and K of B and D (respectively), thus if M exceeds N then G exceeds K, and if (M is) equal (to N then G is also) equal (to K), and if (M is) less (than N then G is also) less (than K) [Def. 5.5]. And G exceeds K. Thus, M also exceeds N. And H does not exceeds L. And M and H are equal multiples of A and E (respectively), and N and L other random equal multiples of B and F (respectively). Thus, A has a greater ratio to B than E (has) to F [Def. 5.7].

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and a third (magnitude) has a greater ratio to a fourth than a fifth (has) to a sixth, then the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta > \epsilon : \zeta$ then $\alpha : \beta > \epsilon : \zeta$.

ιδ′.

Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ δεύτερον τοῦ τετάρτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον.



Πρῶτον γὰρ τὸ Α πρὸς δεύτερον τὸ Β αὐτὸν ἐχέτω λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ, μεῖζον δὲ ἔστω τὸ Α τοῦ Γ· λέγω, ὅτι καὶ τὸ Β τοῦ Δ μεῖζόν ἐστιν.

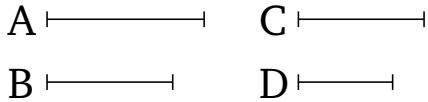
Ἐπεὶ γὰρ τὸ Α τοῦ Γ μεῖζόν ἐστιν, ἄλλο δέ, ὃ ἔτυχεν, [μέγεθος] τὸ Β, τὸ Α ἄρα πρὸς τὸ Β μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Β. ὡς δὲ τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ· καὶ τὸ Γ ἄρα πρὸς τὸ Δ μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Β. πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔχει, ἐκεῖνο ἔλασσόν ἐστιν· ἔλασσον ἄρα τὸ Δ τοῦ Β· ὥστε μεῖζόν ἐστι τὸ Β τοῦ Δ.

Ὁμοίως δὴ δεῖξομεν, ὅτι κἂν ἴσον ᾖ τὸ Α τῷ Γ, ἴσον ἔσται καὶ τὸ Β τῷ Δ, κἂν ἔλασσον ᾖ τὸ Α τοῦ Γ, ἔλασσον ἔσται καὶ τὸ Β τοῦ Δ.

Ἐὰν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ δεύτερον τοῦ τετάρτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖξαι.

Proposition 14†

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, then the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth).



For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D. And let A be greater than C. I say that B is also greater than D.

For since A is greater than C, and B (is) another random [magnitude], A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. And as A (is) to B, so C (is) to D. Thus, C also has a greater ratio to D than C (has) to B. And that (magnitude) to which the same (magnitude) has a greater ratio is the lesser [Prop. 5.10]. Thus, D (is) less than B. Hence, B is greater than D.

So, similarly, we can show that even if A is equal to C then B will also be equal to D, and even if A is less than C then B will also be less than D.

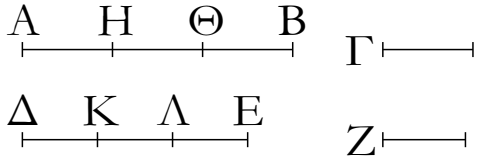
Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, then the second will also be greater than the fourth. And if (the first magnitude is)

equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha \gtreqless \gamma$ as $\beta \gtreqless \delta$.

ιε′.

Τὰ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον ληφθέντα κατάλληλα.



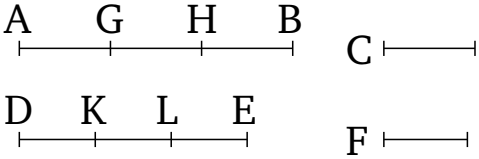
Ἔστω γὰρ ἰσάκις πολλαπλάσιον τὸ ΑΒ τοῦ Γ καὶ το ΔΕ τοῦ Ζ· λέγω, ὅτι ἐστὶν ὡς τὸ Γ πρὸς τὸ Ζ, οὕτως τὸ ΑΒ πρὸς τὸ ΔΕ.

Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΒ τοῦ Γ καὶ τὸ ΔΕ τοῦ Ζ, ὅσα ἄρα ἐστὶν ἐν τῷ ΑΒ μεγέθη ἴσα τῷ Γ, τοσαῦτα καὶ ἐν τῷ ΔΕ ἴσα τῷ Ζ. διῃρήσθω τὸ μὲν ΑΒ εἰς τὰ τῷ Γ ἴσα τὰ ΑΗ, ΗΘ, ΘΒ, τὸ δὲ ΔΕ εἰς τὰ τῷ Ζ ἴσα τὰ ΔΚ, ΚΛ, ΛΕ· ἔσται δὴ ἴσον τὸ πλῆθος τῶν ΑΗ, ΗΘ, ΘΒ τῷ πλήθει τῶν ΔΚ, ΚΛ, ΛΕ. καὶ ἐπεὶ ἴσα ἐστὶ τὰ ΑΗ, ΗΘ, ΘΒ ἀλλήλοις, ἔστι δὲ καὶ τὰ ΔΚ, ΚΛ, ΛΕ ἴσα ἀλλήλοις, ἔστιν ἄρα ὡς τὸ ΑΗ πρὸς τὸ ΔΚ, οὕτως τὸ ΗΘ πρὸς τὸ ΚΛ, καὶ τὸ ΘΒ πρὸς τὸ ΛΕ. ἔσται ἄρα καὶ ὡς ἓν τῶν ἡγουμένων πρὸς ἓν τῶν ἑπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἑπόμενα· ἔστιν ἄρα ὡς τὸ ΑΗ πρὸς τὸ ΔΚ, οὕτως τὸ ΑΒ πρὸς τὸ ΔΕ. ἴσον δὲ τὸ μὲν ΑΗ τῷ Γ, τὸ δὲ ΔΚ τῷ Ζ· ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Ζ οὕτως τὸ ΑΒ πρὸς τὸ ΔΕ.

Τὰ ἄρα μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

Proposition 15†

Parts have the same ratio as similar multiples, taken in corresponding order.



For let AB and DE be equal multiples of C and F (respectively). I say that as C is to F, so AB (is) to DE.

For since AB and DE are equal multiples of C and F (respectively), thus as many magnitudes as there are in AB equal to C, so many (are there) also in DE equal to F. Let AB have been divided into (magnitudes) AG, GH, HB, equal to C, and DE into (magnitudes) DK, KL, LE, equal to F. So, the number of (magnitudes) AG, GH, HB will equal the number of (magnitudes) DK, KL, LE. And since AG, GH, HB are equal to one another, and DK, KL, LE are also equal to one another, thus as AG is to DK, so GH (is) to KL, and HB to LE [Prop. 5.7]. And, thus (for proportional magnitudes), as one of the leading (magnitudes) will be to one of the following, so all of the leading (magnitudes will be) to all of the following [Prop. 5.12]. Thus, as AG is to DK, so AB (is) to DE. And AG is equal to C, and DK to F. Thus, as C is to F, so AB (is) to DE.

Thus, parts have the same ratio as similar multiples, taken in corresponding order. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that $\alpha : \beta :: m\,\alpha : m\,\beta$.

ιϛ′.

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾖ, καὶ ἐναλλὰξ ἀνάλογον ἔσται.

Ἔστω τέσσαρα μεγέθη ἀνάλογον τὰ Α, Β, Γ, Δ, ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ· λέγω, ὅτι καὶ ἐναλλὰξ [ἀνάλογον] ἔσται, ὡς τὸ Α πρὸς τὸ Γ, οὕτως τὸ Β πρὸς τὸ Δ.

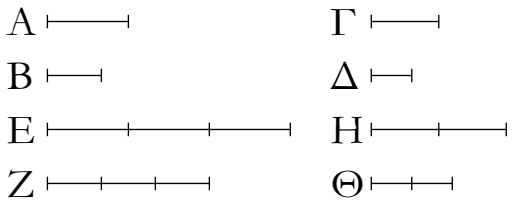
Εἰλήφθω γὰρ τῶν μὲν Α, Β ἰσάκις πολλαπλάσια τὰ Ε, Ζ, τῶν δὲ Γ, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Η, Θ.

Proposition 16†

If four magnitudes are proportional then they will also be proportional alternately.

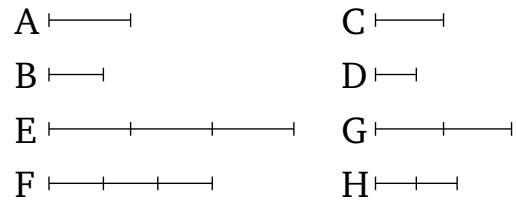
Let A, B, C and D be four proportional magnitudes, (such that) as A (is) to B, so C (is) to D. I say that they will also be [proportional] alternately, (so that) as A (is) to C, so B (is) to D.

For let the equal multiples E and F have been taken of A and B (respectively), and the other random equal multiples G and H of C and D (respectively).



Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ Ε τοῦ Α καὶ τὸ Ζ τοῦ Β, τὰ δὲ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον, ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Ε πρὸς τὸ Ζ. ὡς δὲ τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ· καὶ ὡς ἄρα τὸ Γ πρὸς τὸ Δ, οὕτως τὸ Ε πρὸς τὸ Ζ. πάλιν, ἐπεὶ τὰ Η, Θ τῶν Γ, Δ ἰσάκις ἐστὶ πολλαπλάσια, ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Δ, οὕτως τὸ Η πρὸς τὸ Θ. ὡς δὲ τὸ Γ πρὸς τὸ Δ, [οὕτως] τὸ Ε πρὸς τὸ Ζ· καὶ ὡς ἄρα τὸ Ε πρὸς τὸ Ζ, οὕτως τὸ Η πρὸς τὸ Θ. ἐὰν δὲ τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ δὲ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ δεύτερον τοῦ τετάρτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον. εἰ ἄρα ὑπερέχει τὸ Ε τοῦ Η, ὑπερέχει καὶ τὸ Ζ τοῦ Θ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν Ε, Ζ τῶν Α, Β ἰσάκις πολλαπλάσια, τὰ δὲ Η, Θ τῶν Γ, Δ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Γ, οὕτως τὸ Β πρὸς τὸ Δ.

Ἐὰν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾖ, καὶ ἐναλλὰξ ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.



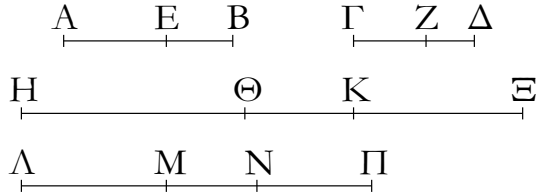
And since E and F are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A is to B, so E (is) to F. But as A (is) to B, so C (is) to D. And, thus, as C (is) to D, so E (is) to F [Prop. 5.11]. Again, since G and H are equal multiples of C and D (respectively), thus as C is to D, so G (is) to H [Prop. 5.15]. But as C (is) to D, [so] E (is) to F. And, thus, as E (is) to F, so G (is) to H [Prop. 5.11]. And if four magnitudes are proportional, and the first is greater than the third then the second will also be greater than the fourth, and if (the first is) equal (to the third then the second will also be) equal (to the fourth), and if (the first is) less (than the third then the second will also be) less (than the fourth) [Prop. 5.14]. Thus, if E exceeds G then F also exceeds H, and if (E is) equal (to G then F is also) equal (to H), and if (E is) less (than G then F is also) less (than H). And E and F are equal multiples of A and B (respectively), and G and H other random equal multiples of C and D (respectively). Thus, as A is to C, so B (is) to D [Def. 5.5].

Thus, if four magnitudes are proportional then they will also be proportional alternately. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \gamma :: \beta : \delta$.

ιζ′.

Ἐὰν συγκείμενα μεγέθη ἀνάλογον ᾖ, καὶ διαιρεθέντα ἀνάλογον ἔσται.



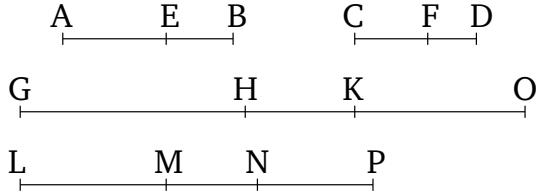
Ἔστω συγκείμενα μεγέθη ἀνάλογον τὰ ΑΒ, ΒΕ, ΓΔ, ΔΖ, ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς τὸ ΔΖ· λέγω, ὅτι καὶ διαιρεθέντα ἀνάλογον ἔσται, ὡς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΖ πρὸς τὸ ΔΖ.

Εἰλήφθω γὰρ τῶν μὲν ΑΕ, ΕΒ, ΓΖ, ΖΔ ἰσάκις πολλαπλάσια τὰ ΗΘ, ΘΚ, ΛΜ, ΜΝ, τῶν δὲ ΕΒ, ΖΔ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ ΚΞ, ΝΠ.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΘΚ τοῦ ΕΒ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ

Proposition 17†

If composed magnitudes are proportional then they will also be proportional (when) separarted.



Let AB, BE, CD, and DF be composed magnitudes (which are) proportional, (so that) as AB (is) to BE, so CD (is) to DF. I say that they will also be proportional (when) separated, (so that) as AE (is) to EB, so CF (is) to DF.

For let the equal multiples GH, HK, LM, and MN have been taken of AE, EB, CF, and FD (respectively), and the other random equal multiples KO and NP of EB and FD (respectively).

ΑΕ καὶ τὸ ΗΚ τοῦ ΑΒ. ἰσάκις δέ ἐστι πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΛΜ τοῦ ΓΖ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΛΜ τοῦ ΓΖ. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΜΝ τοῦ ΖΔ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΛΝ τοῦ ΓΔ. ἰσάκις δὲ ἦν πολλαπλάσιον τὸ ΛΜ τοῦ ΓΖ καὶ τὸ ΗΚ τοῦ ΑΒ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΛΝ τοῦ ΓΔ. τὰ ΗΚ, ΛΝ ἄρα τῶν ΑΒ, ΓΔ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλασίον τὸ ΘΚ τοῦ ΕΒ καὶ τὸ ΜΝ τοῦ ΖΔ, ἔστι δὲ καὶ τὸ ΚΞ τοῦ ΕΒ ἰσάκις πολλαπλάσιον καὶ τὸ ΝΠ τοῦ ΖΔ, καὶ συντεθὲν τὸ ΘΞ τοῦ ΕΒ ἰσάκις ἐστὶ πολλαπλάσιον καὶ τὸ ΜΠ τοῦ ΖΔ. καὶ ἐπεί ἐστιν ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς τὸ ΔΖ, καὶ εἴληπται τῶν μὲν ΑΒ, ΓΔ ἰσάκις πολλαπλάσια τὰ ΗΚ, ΛΝ, τῶν δὲ ΕΒ, ΖΔ ἰσάκις πολλαπλάσια τὰ ΘΞ, ΜΠ, εἰ ἄρα ὑπερέχει τὸ ΗΚ τοῦ ΘΞ, ὑπερέχει καὶ τὸ ΛΝ τοῦ ΜΠ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὑπερεχέτω δὴ τὸ ΗΚ τοῦ ΘΞ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΘΚ ὑπερέχει ἄρα καὶ τὸ ΗΘ τοῦ ΚΞ. ἀλλα εἰ ὑπερεῖχε τὸ ΗΚ τοῦ ΘΞ ὑπερεῖχε καὶ τὸ ΛΝ τοῦ ΜΠ· ὑπερέχει ἄρα καὶ τὸ ΛΝ τοῦ ΜΠ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΜΝ ὑπερέχει καὶ τὸ ΛΜ τοῦ ΝΠ· ὥστε εἰ ὑπερέχει τὸ ΗΘ τοῦ ΚΞ, ὑπερέχει καὶ τὸ ΛΜ τοῦ ΝΠ. ὁμοίως δὴ δεῖξομεν, ὅτι κἂν ἴσον ᾖ τὸ ΗΘ τῷ ΚΞ, ἴσον ἔσται καὶ τὸ ΛΜ τῷ ΝΠ, κἂν ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν ΗΘ, ΛΜ τῶν ΑΕ, ΓΖ ἰσάκις πολλαπλάσια, τὰ δὲ ΚΞ, ΝΠ τῶν ΕΒ, ΖΔ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΖ πρὸς τὸ ΖΔ.

Ἐὰν ἄρα συγκείμενα μεγέθη ἀνάλογον ᾖ, καὶ διαιρεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

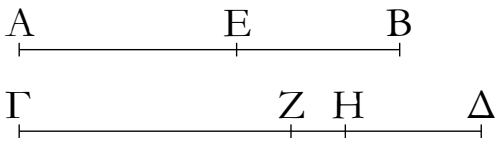
And since GH and HK are equal multiples of AE and EB (respectively), GH and GK are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. But GH and LM are equal multiples of AE and CF (respectively). Thus, GK and LM are equal multiples of AB and CF (respectively). Again, since LM and MN are equal multiples of CF and FD (respectively), LM and LN are thus equal multiples of CF and CD (respectively) [Prop. 5.1]. And LM and GK were equal multiples of CF and AB (respectively). Thus, GK and LN are equal multiples of AB and CD (respectively). Thus, GK, LN are equal multiples of AB, CD. Again, since HK and MN are equal multiples of EB and FD (respectively), and KO and NP are also equal multiples of EB and FD (respectively), then, added together, HO and MP are also equal multiples of EB and FD (respectively) [Prop. 5.2]. And since as AB (is) to BE, so CD (is) to DF, and the equal multiples GK, LN have been taken of AB, CD, and the equal multiples HO, MP of EB, FD, thus if GK exceeds HO then LN also exceeds MP, and if (GK is) equal (to HO then LN is also) equal (to MP), and if (GK is) less (than HO then LN is also) less (than MP) [Def. 5.5]. So let GK exceed HO, and thus, HK being taken away from both, GH exceeds KO. But (we saw that) if GK was exceeding HO then LN was also exceeding MP. Thus, LN also exceeds MP, and, MN being taken away from both, LM also exceeds NP. Hence, if GH exceeds KO then LM also exceeds NP. So, similarly, we can show that even if GH is equal to KO then LM will also be equal to NP, and even if (GH is) less (than KO then LM will also be) less (than NP). And GH, LM are equal multiples of AE, CF, and KO, NP other random equal multiples of EB, FD. Thus, as AE is to EB, so CF (is) to FD [Def. 5.5].

Thus, if composed magnitudes are proportional then they will also be proportional (when) separarted. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha + \beta : \beta :: \gamma + \delta : \delta$ then $\alpha : \beta :: \gamma : \delta$.

ιη′.

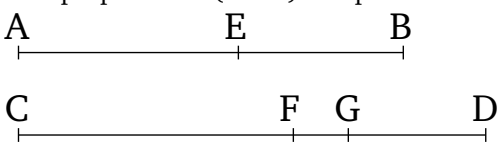
Ἐὰν διῃρημένα μεγέθη ἀνάλογον ᾖ, καὶ συντεθέντα ἀνάλογον ἔσται.



Ἔστω διῃρημένα μεγέθη ἀνάλογον τὰ ΑΕ, ΕΒ, ΓΖ, ΖΔ, ὡς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΖ πρὸς τὸ ΖΔ· λέγω, ὅτι καὶ συντεθέντα ἀνάλογον ἔσται, ὡς τὸ ΑΒ πρὸς τὸ ΒΕ,

Proposition 18†

If separated magnitudes are proportional then they will also be proportional (when) composed.



Let AE, EB, CF, and FD be separated magnitudes (which are) proportional, (so that) as AE (is) to EB, so CF (is) to FD. I say that they will also be proportional

οὕτως τὸ ΓΔ πρὸς τὸ ΖΔ.

Εἰ γὰρ μή ἐστὶν ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς τὸ ΔΖ, ἔσται ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ ἤτοι πρὸς ἔλασσόν τι τοῦ ΔΖ ἢ πρὸς μεῖζον.

Ἔστω πρότερον πρὸς ἔλασσον τὸ ΔΗ. καὶ ἐπεί ἐστιν ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς τὸ ΔΗ, συγκείμενα μεγέθη ἀνάλογόν ἐστιν· ὥστε καὶ διαιρεθέντα ἀνάλογον ἔσται. ἔστιν ἄρα ὡς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΗ πρὸς τὸ ΗΔ. ὑπόκειται δὲ καὶ ὡς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΖ πρὸς τὸ ΖΔ. καὶ ὡς ἄρα τὸ ΓΗ πρὸς τὸ ΗΔ, οὕτως τὸ ΓΖ πρὸς τὸ ΖΔ. μεῖζον δὲ τὸ πρῶτον τὸ ΓΗ τοῦ τρίτου τοῦ ΓΖ· μεῖζον ἄρα καὶ τὸ δεύτερον τὸ ΗΔ τοῦ τετάρτου τοῦ ΖΔ. ἀλλὰ καὶ ἔλαττον· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα ἐστὶν ὡς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς ἔλασσον τοῦ ΖΔ. ὁμοίως δὴ δείξομεν, ὅτι οὐδὲ πρὸς μεῖζον· πρὸς αὐτὸ ἄρα.

Ἐὰν ἄρα διῃρημένα μεγέθη ἀνάλογον ᾖ, καὶ συντεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

(when) composed, (so that) as AB (is) to BE, so CD (is) to FD.

For if (it is) not (the case that) as AB is to BE, so CD (is) to FD, then it will surely be (the case that) as AB (is) to BE, so CD is either to some (magnitude) less than DF, or (some magnitude) greater (than DF).‡

Let it, first of all, be to (some magnitude) less (than DF), (namely) DG. And since composed magnitudes are proportional, (so that) as AB is to BE, so CD (is) to DG, they will thus also be proportional (when) separated [Prop. 5.17]. Thus, as AE is to EB, so CG (is) to GD. But it was also assumed that as AE (is) to EB, so CF (is) to FD. Thus, (it is) also (the case that) as CG (is) to GD, so CF (is) to FD [Prop. 5.11]. And the first (magnitude) CG (is) greater than the third CF. Thus, the second (magnitude) GD (is) also greater than the fourth FD [Prop. 5.14]. But (it is) also less. The very thing is impossible. Thus, (it is) not (the case that) as AB is to BE, so CD (is) to less than FD. Similarly, we can show that neither (is it the case) to greater (than FD). Thus, (it is the case) to the same (as FD).

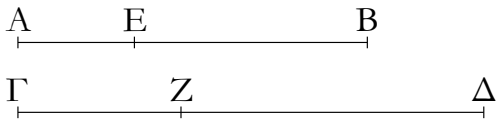
Thus, if separated magnitudes are proportional then they will also be proportional (when) composed. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha + \beta : \beta :: \gamma + \delta : \delta$.

‡ Here, Euclid assumes, without proof, that a fourth magnitude proportional to three given magnitudes can always be found.

ιθ′.

Ἐὰν ᾖ ὡς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθὲν πρὸς ἀφαιρεθέν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὡς ὅλον πρὸς ὅλον.



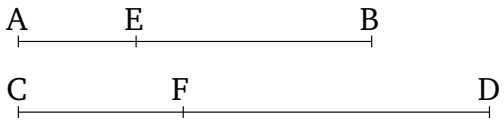
Ἔστω γὰρ ὡς ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ, οὕτως ἀφαιρεθὲν τὸ ΑΕ πρὸς ἀφαιρεθὲν τὸ ΓΖ· λέγω, ὅτι καὶ λοιπὸν τὸ ΕΒ πρὸς λοιπὸν τὸ ΖΔ ἔσται ὡς ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ.

Ἐπεὶ γάρ ἐστιν ὡς τὸ ΑΒ πρὸς τὸ ΓΔ, οὕτως τὸ ΑΕ πρὸς τὸ ΓΖ, καὶ ἐναλλὰξ ὡς τὸ ΒΑ πρὸς τὸ ΑΕ, οὕτως τὸ ΔΓ πρὸς τὸ ΓΖ. καὶ ἐπεὶ συγκείμενα μεγέθη ἀνάλογόν ἐστιν, καὶ διαιρεθέντα ἀνάλογον ἔσται, ὡς τὸ ΒΕ πρὸς τὸ ΕΑ, οὕτως τὸ ΔΖ πρὸς τὸ ΓΖ· καὶ ἐναλλάξ, ὡς τὸ ΒΕ πρὸς τὸ ΔΖ, οὕτως τὸ ΕΑ πρὸς τὸ ΖΓ. ὡς δὲ τὸ ΑΕ πρὸς τὸ ΓΖ, οὕτως ὑπόκειται ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ. καὶ λοιπὸν ἄρα τὸ ΕΒ πρὸς λοιπὸν τὸ ΖΔ ἔσται ὡς ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ.

Ἐὰν ἄρα ᾖ ὡς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθὲν πρὸς

Proposition 19†

If as the whole is to the whole so the (part) taken away is to the (part) taken away then the remainder to the remainder will also be as the whole (is) to the whole.



For let the whole AB be to the whole CD as the (part) taken away AE (is) to the (part) taken away CF. I say that the remainder EB to the remainder FD will also be as the whole AB (is) to the whole CD.

For since as AB is to CD, so AE (is) to CF, (it is) also (the case), alternately, (that) as BA (is) to AE, so DC (is) to CF [Prop. 5.16]. And since composed magnitudes are proportional then they will also be proportional (when) separated, (so that) as BE (is) to EA, so DF (is) to CF [Prop. 5.17]. Also, alternately, as BE (is) to DF, so EA (is) to FC [Prop. 5.16]. And it was assumed that as AE (is) to CF, so the whole AB (is) to the whole CD. And, thus, as the remainder EB (is) to the remainder FD, so the whole AB will be to the whole CD.

ἀφαιρεθέν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὡς ὅλον πρὸς ὅλον [ὅπερ ἔδει δεῖξαι].

[Καὶ ἐπεὶ ἐδείχθη ὡς τὸ ΑΒ πρὸς τὸ ΓΔ, οὕτως τὸ ΕΒ πρὸς τὸ ΖΔ, καὶ ἐναλλὰξ ὡς τὸ ΑΒ πρὸς τὸ ΒΕ οὕτως τὸ ΓΔ πρὸς τὸ ΖΔ, συγκείμενα ἄρα μεγέθη ἀνάλογόν ἐστιν· ἐδείχθη δὲ ὡς τὸ ΒΑ πρὸς τὸ ΑΕ, οὕτως τὸ ΔΓ πρὸς τὸ ΓΖ· καί ἐστιν ἀναστρέψαντι].

Thus, if as the whole is to the whole so the (part) taken away is to the (part) taken away then the remainder to the remainder will also be as the whole (is) to the whole. [(Which is) the very thing it was required to show.]

[And since it was shown (that) as AB (is) to CD, so EB (is) to FD, (it is) also (the case), alternately, (that) as AB (is) to BE, so CD (is) to FD. Thus, composed magnitudes are proportional. And it was shown (that) as BA (is) to AE, so DC (is) to CF. And (the latter) is converted (from the former).]

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐὰν συγκείμενα μεγέθη ἀνάλογον ᾖ, καὶ ἀναστρέψαντι ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

Corollary‡

So (it is) clear, from this, that if composed magnitudes are proportional then they will also be proportional (when) converted. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \beta :: \alpha - \gamma : \beta - \delta$.

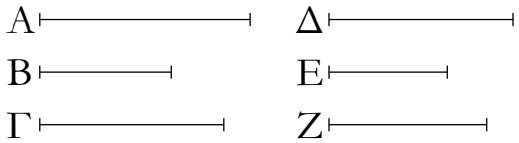
‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \alpha - \beta :: \gamma : \gamma - \delta$.

κ′.

Ἐὰν ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι᾽ ἴσου δὲ τὸ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ τέταρτον τοῦ ἕκτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον.

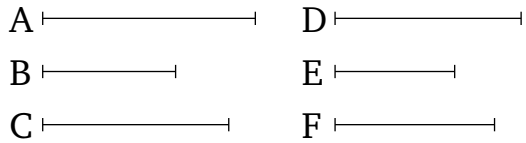
Proposition 20†

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third then the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Ἔστω τρία μεγέθη τὰ Α, Β, Γ, καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὡς μὲν τὸ Α πρὸς τὸ Β, οὕτως τὸ Δ πρὸς τὸ Ε, ὡς δὲ τὸ Β πρὸς τὸ Γ, οὕτως τὸ Ε πρὸς τὸ Ζ, δι᾽ ἴσου δὲ μεῖζον ἔστω τὸ Α τοῦ Γ· λέγω, ὅτι καὶ τὸ Δ τοῦ Ζ μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μεῖζόν ἐστι τὸ Α τοῦ Γ, ἄλλο δέ τι τὸ Β, τὸ δὲ μεῖζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔχει ἤπερ τὸ ἔλαττον, τὸ Α ἄρα πρὸς τὸ Β μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Β. ἀλλ᾽ ὡς μὲν τὸ Α πρὸς τὸ Β [οὕτως] τὸ Δ πρὸς τὸ Ε, ὡς δὲ τὸ Γ πρὸς τὸ Β, ἀνάπαλιν οὕτως τὸ Ζ πρὸς τὸ Ε· καὶ τὸ Δ ἄρα πρὸς τὸ Ε μείζονα λόγον ἔχει ἤπερ τὸ Ζ πρὸς τὸ Ε. τῶν δὲ πρὸς τὸ αὐτὸ λόγον ἐχόντων τὸ μείζονα λόγον ἔχον μεῖζόν ἐστιν. μεῖζον ἄρα τὸ Δ τοῦ Ζ. ὁμοίως δὴ δείξομεν, ὅτι κἂν ἴσον ᾖ τὸ Α τῷ Γ, ἴσον ἔσται καὶ τὸ Δ τῷ Ζ, κἂν



Let A, B, and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, (so that) as A (is) to B, so D (is) to E, and as B (is) to C, so E (is) to F. And let A be greater than C, via equality. I say that D will also be greater than F. And if (A is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C, and B some other (magnitude), and the greater (magnitude) has a greater ratio than the lesser to the same (magnitude) [Prop. 5.8], A thus has a greater ratio to B than C (has) to B. But as A (is) to B, [so] D (is) to E. And, inversely, as C (is) to B, so F (is) to E [Prop. 5.7 corr.]. Thus, D also has a greater ratio to E than F (has) to E [Prop. 5.13]. And for (mag-

ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι᾽ ἴσου δὲ τὸ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ τέταρτον τοῦ ἕκτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖξαι.

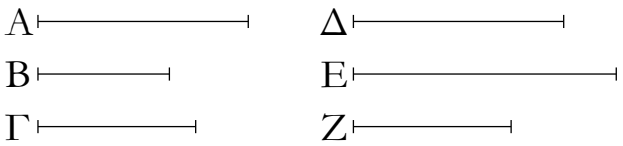
nitudes) having a ratio to the same (magnitude), that having the greater ratio is greater [Prop. 5.10]. Thus, D (is) greater than F. Similarly, we can show that even if A is equal to C then D will also be equal to F, and even if (A is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third, then the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And (if the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \delta : \epsilon$ and $\beta : \gamma :: \epsilon : \zeta$ then $\alpha \gtreqless \gamma$ as $\delta \gtreqless \zeta$.

κα′.

Ἐὰν ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ᾖ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι᾽ ἴσου δὲ τὸ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ τέταρτον τοῦ ἕκτου μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον.



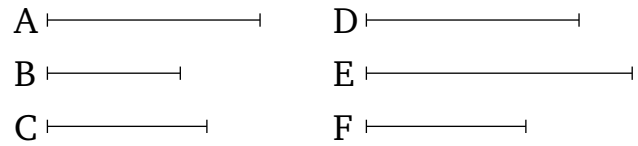
Ἔστω τρία μεγέθη τὰ Α, Β, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὡς μὲν τὸ Α πρὸς τὸ Β, οὕτως τὸ Ε πρὸς τὸ Ζ, ὡς δὲ τὸ Β πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ Ε, δι᾽ ἴσου δὲ τὸ Α τοῦ Γ μεῖζον ἔστω· λέγω, ὅτι καὶ τὸ Δ τοῦ Ζ μεῖζον ἔσται, κἂν ἴσον, ἴσον, κἂν ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μεῖζόν ἐστι τὸ Α τοῦ Γ, ἄλλο δέ τι τὸ Β, τὸ Α ἄρα πρὸς τὸ Β μείζονα λόγον ἔχει ἤπερ τὸ Γ πρὸς τὸ Β. ἀλλ᾽ ὡς μὲν τὸ Α πρὸς τὸ Β, οὕτως τὸ Ε πρὸς τὸ Ζ, ὡς δὲ τὸ Γ πρὸς τὸ Β, ἀνάπαλιν οὕτως τὸ Ε πρὸς τὸ Δ. καὶ τὸ Ε ἄρα πρὸς τὸ Ζ μείζονα λόγον ἔχει ἤπερ τὸ Ε πρὸς τὸ Δ. πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔχει, ἐκεῖνο ἔλασσόν ἐστιν· ἔλασσον ἄρα ἐστὶ τὸ Ζ τοῦ Δ· μεῖζον ἄρα ἐστὶ τὸ Δ τοῦ Ζ. ὁμοίως δὴ δείξομεν, ὅτι κἂν ἴσον ᾖ τὸ Α τῷ Γ, ἴσον ἔσται καὶ τὸ Δ τῷ Ζ, κἂν ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ᾖ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι᾽ ἴσου δὲ τὸ πρῶτον τοῦ τρίτου μεῖζον ᾖ, καὶ τὸ τέταρτον τοῦ ἕκτου μεῖζον ἔσται, κἂν ἴσον,

Proposition 21†

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third then the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Let A, B, and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B, so E (is) to F, and as B (is) to C, so D (is) to E. And let A be greater than C, via equality. I say that D will also be greater than F. And if (A is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C, and B some other (magnitude), A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. But as A (is) to B, so E (is) to F. And, inversely, as C (is) to B, so E (is) to D [Prop. 5.7 corr.]. Thus, E also has a greater ratio to F than E (has) to D [Prop. 5.13]. And that (magnitude) to which the same (magnitude) has a greater ratio is (the) lesser (magnitude) [Prop. 5.10]. Thus, F is less than D. Thus, D is greater than F. Similarly, we can show that even if A is equal to C then D will also be equal to F, and even if (A is) less (than C then D will also be) less (than F).

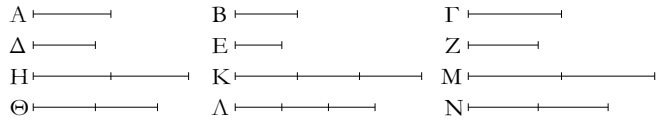
ἴσον, κἂν ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖξαι.

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third then the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha \gtreqless \gamma$ as $\delta \gtreqless \zeta$.

κβ′.

Ἐὰν ᾖ ὁποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι᾽ ἴσου ἐν τῷ αὐτῷ λόγῳ ἔσται.



Ἔστω ὁποσαοῦν μεγέθη τὰ Α, Β, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, Ε, Ζ, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὡς μὲν τὸ Α πρὸς τὸ Β, οὕτως τὸ Δ πρὸς τὸ Ε, ὡς δὲ τὸ Β πρὸς τὸ Γ, οὕτως τὸ Ε πρὸς τὸ Ζ· λέγω, ὅτι καὶ δι᾽ ἴσου ἐν τῷ αὐτῳ λόγῳ ἔσται.

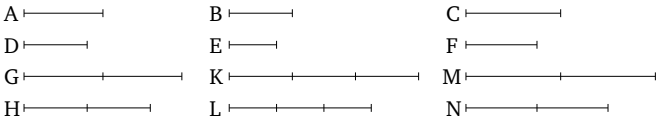
Εἰλήφθω γὰρ τῶν μὲν Α, Δ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Ε ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Κ, Λ, καὶ ἔτι τῶν Γ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Μ, Ν.

Καὶ ἐπεί ἐστιν ὡς τὸ Α πρὸς τὸ Β, οὕτως τὸ Δ πρὸς τὸ Ε, καὶ εἵληπται τῶν μὲν Α, Δ ἰσάκις πολλαπλάσια τὰ Η, Θ, τῶν δὲ Β, Ε ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Κ, Λ, ἔστιν ἄρα ὡς τὸ Η πρὸς τὸ Κ, οὕτως τὸ Θ πρὸς τὸ Λ. δἰα τὰ αὐτὰ δὴ καὶ ὡς τὸ Κ πρὸς τὸ Μ, οὕτως τὸ Λ πρὸς τὸ Ν. ἐπεὶ οὖν τρία μεγέθη ἐστὶ τὰ Η, Κ, Μ, καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Θ, Λ, Ν, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι᾽ ἴσου ἄρα, εἰ ὑπερέχει τὸ Η τοῦ Μ, ὑπερέχει καὶ τὸ Θ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν Η, Θ τῶν Α, Δ ἰσάκις πολλαπλάσια, τὰ δὲ Μ, Ν τῶν Γ, Ζ ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια. ἔστιν ἄρα ὡς τὸ Α πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ Ζ.

Ἐὰν ἄρα ᾖ ὁποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ δι᾽ ἴσου ἐν τῷ αὐτῷ λόγῳ ἔσται· ὅπερ ἔδει δεῖξαι.

Proposition 22†

If there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, then they will also be in the same ratio via equality.



Let there be any number of magnitudes whatsoever, A, B, C, and (some) other (magnitudes), D, E, F, of equal number to them, (which are) in the same ratio taken two by two, (so that) as A (is) to B, so D (is) to E, and as B (is) to C, so E (is) to F. I say that they will also be in the same ratio via equality. (That is, as A is to C, so D is to F.)

For let the equal multiples G and H have been taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), and the yet other random equal multiples M and N of C and F (respectively).

And since as A is to B, so D (is) to E, and the equal multiples G and H have been taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), thus as G is to K, so H (is) to L [Prop. 5.4]. And, so, for the same (reasons), as K (is) to M, so L (is) to N. Therefore, since G, K, and M are three magnitudes, and H, L, and N other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, thus, via equality, if G exceeds M then H also exceeds N, and if (G is) equal (to M then H is also) equal (to N), and if (G is) less (than M then H is also) less (than N) [Prop. 5.20]. And G and H are equal multiples of A and D (respectively), and M and N other random equal multiples of C and F (respectively). Thus, as A is to C, so D (is) to F [Def. 5.5].

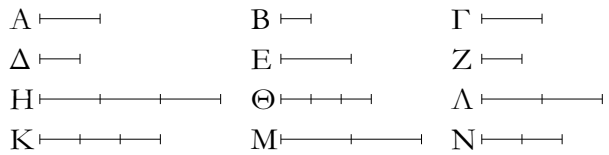
Thus, if there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by

two, then they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \zeta : \eta$ and $\gamma : \delta :: \eta : \theta$ then $\alpha : \delta :: \epsilon : \theta$.

κγ′.

Ἐὰν ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ᾖ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι᾽ ἴσου ἐν τῷ αὐτῷ λόγῳ ἔσται.



Ἔστω τρία μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ τὰ Δ, E, Z, ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὡς μὲν τὸ A πρὸς τὸ B, οὕτως τὸ E πρὸς τὸ Z, ὡς δὲ τὸ B πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ E· λέγω, ὅτι ἐστὶν ὡς τὸ A πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ Z.

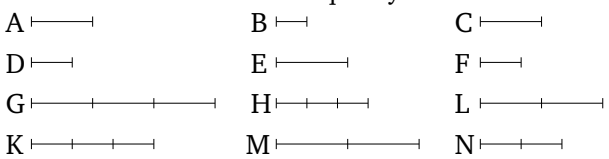
Εἰλήφθω τῶν μὲν A, B, Δ ἰσάκις πολλαπλάσια τὰ H, Θ, K, τῶν δὲ Γ, E, Z ἄλλα, ἃ ἔτυχεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσια τὰ H, Θ τῶν A, B, τὰ δὲ μέρη τοὶς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον, ἔστιν ἄρα ὡς τὸ A πρὸς τὸ B, οὕτως τὸ H πρὸς τὸ Θ. διὰ τὰ αὐτὰ δὴ καὶ ὡς τὸ E πρὸς τὸ Z, οὕτως τὸ M πρὸς τὸ N· καί ἐστιν ὡς τὸ A πρὸς τὸ B, οὕτως τὸ E πρὸς τὸ Z· καὶ ὡς ἄρα τὸ H πρὸς τὸ Θ, οὕτως τὸ M πρὸς τὸ N. καὶ ἐπεί ἐστιν ὡς τὸ B πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ E, καὶ ἐναλλὰξ ὡς τὸ B πρὸς τὸ Δ, οὕτως τὸ Γ πρὸς τὸ E. καὶ ἐπεὶ τὰ Θ, K τῶν B, Δ ἰσάκις ἐστὶ πολλαπλάσια, τὰ δὲ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον, ἔστιν ἄρα ὡς τὸ B πρὸς τὸ Δ, οὕτως τὸ Θ πρὸς τὸ K. ἀλλ᾽ ὡς τὸ B πρὸς τὸ Δ, οὕτως τὸ Γ πρὸς τὸ E· καὶ ὡς ἄρα τὸ Θ πρὸς τὸ K, οὕτως τὸ Γ πρὸς τὸ E. πάλιν, ἐπεὶ τὰ Λ, M τῶν Γ, E ἰσάκις ἐστι πολλαπλάσια, ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ E, οὕτως τὸ Λ πρὸς τὸ M. ἀλλ᾽ ὡς τὸ Γ πρὸς τὸ E, οὕτως τὸ Θ πρὸς τὸ K· καὶ ὡς ἄρα τὸ Θ πρὸς τὸ K, οὕτως τὸ Λ πρὸς τὸ M, καὶ ἐναλλὰξ ὡς τὸ Θ πρὸς τὸ Λ, τὸ K πρὸς τὸ M. ἐδείχθη δὲ καὶ ὡς τὸ H πρὸς τὸ Θ, οὕτως τὸ M πρὸς τὸ N. ἐπεὶ οὖν τρία μεγέθη ἐστὶ τὰ H, Θ, Λ, καὶ ἄλλα αὐτοις ἴσα τὸ πλῆθος τὰ K, M, N σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καί ἐστιν αὐτῶν τεταραγμένη ἡ ἀναλογία, δι᾽ ἴσου ἄρα, εἰ ὑπερέχει τὸ H τοῦ Λ, ὑπερέχει καὶ τὸ K τοῦ N, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καί ἐστι τὰ μὲν H, K τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν Γ, Z. ἔστιν ἄρα ὡς τὸ A πρὸς τὸ Γ, οὕτως τὸ Δ πρὸς τὸ Z.

Ἐὰν ἄρα ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ᾖ δὲ τεταραγμένη

Proposition 23†

If there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, then they will also be in the same ratio via equality.



Let A, B, and C be three magnitudes, and D, E and F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B, so E (is) to F, and as B (is) to C, so D (is) to E. I say that as A is to C, so D (is) to F.

Let the equal multiples G, H, and K have been taken of A, B, and D (respectively), and the other random equal multiples L, M, and N of C, E, and F (respectively).

And since G and H are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A (is) to B, so G (is) to H. And, so, for the same (reasons), as E (is) to F, so M (is) to N. And as A is to B, so E (is) to F. And, thus, as G (is) to H, so M (is) to N [Prop. 5.11]. And since as B is to C, so D (is) to E, also, alternately, as B (is) to D, so C (is) to E [Prop. 5.16]. And since H and K are equal multiples of B and D (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as B is to D, so H (is) to K. But, as B (is) to D, so C (is) to E. And, thus, as H (is) to K, so C (is) to E [Prop. 5.11]. Again, since L and M are equal multiples of C and E (respectively), thus as C is to E, so L (is) to M [Prop. 5.15]. But, as C (is) to E, so H (is) to K. And, thus, as H (is) to K, so L (is) to M [Prop. 5.11]. Also, alternately, as H (is) to L, so K (is) to M [Prop. 5.16]. And it was also shown (that) as G (is) to H, so M (is) to N. Therefore, since G, H, and L are three magnitudes, and K, M, and N other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, and their proportion is perturbed, thus, via equality, if G exceeds L then K also exceeds N, and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N) [Prop. 5.21]. And G and K are equal multiples of A and D (respectively), and L and N of C and

αὐτῶν ἡ ἀναλογία, καὶ δι᾽ ἴσου ἐν τῷ αὐτῷ λόγῳ ἔσται· ὅπερ ἔδει δεῖξαι.

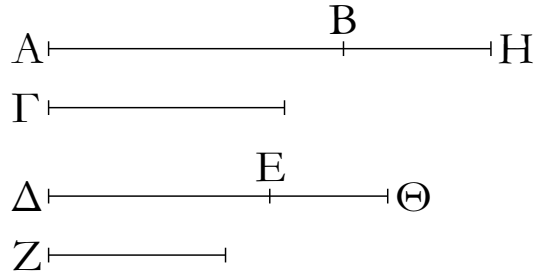
F (respectively). Thus, as A (is) to C, so D (is) to F [Def. 5.5].

Thus, if there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, then they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha : \gamma :: \delta : \zeta$.

κδ′.

Ἐὰν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔχῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἕξει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον.



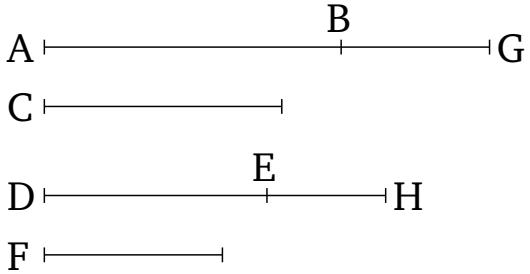
Πρῶτον γὰρ τὸ ΑΒ πρὸς δεύρερον τὸ Γ τὸν αὐτὸν ἐχέτω λόγον καὶ τρίτον τὸ ΔΕ πρὸς τέταρτον τὸ Ζ, ἐχέτω δὲ καὶ πέμπτον τὸ ΒΗ πρὸς δεύτερον τὸ Γ τὸν αὐτὸν λόγον καὶ ἕκτον τὸ ΕΘ πρὸς τέταρτον τὸ Ζ· λέγω, ὅτι καὶ συντεθὲν πρῶτον καὶ πέμπτον τὸ ΑΗ πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἕξει λόγον, καὶ τρίτον καὶ ἕκτον τὸ ΔΘ πρὸς τέταρτον τὸ Ζ.

Ἐπεὶ γάρ ἐστιν ὡς τὸ ΒΗ πρὸς τὸ Γ, οὕτως τὸ ΕΘ πρὸς τὸ Ζ, ἀνάπαλιν ἄρα ὡς τὸ Γ πρὸς τὸ ΒΗ, οὕτως τὸ Ζ πρὸς τὸ ΕΘ. ἐπεὶ οὖν ἐστιν ὡς τὸ ΑΒ πρὸς τὸ Γ, οὕτως τὸ ΔΕ πρὸς τὸ Ζ, ὡς δὲ τὸ Γ πρὸς τὸ ΒΗ, οὕτως τὸ Ζ πρὸς τὸ ΕΘ, δι᾽ ἴσου ἄρα ἐστὶν ὡς τὸ ΑΒ πρὸς τὸ ΒΗ, οὕτως τὸ ΔΕ πρὸς τὸ ΕΘ. καὶ ἐπεὶ διῃρημένα μεγέθη ἀνάλογόν ἐστιν, καὶ συντεθέντα ἀνάλογον ἔσται· ἔστιν ἄρα ὡς τὸ ΑΗ πρὸς τὸ ΗΒ, οὕτως τὸ ΔΘ πρὸς τὸ ΘΕ. ἔστι δὲ καὶ ὡς τὸ ΒΗ πρὸς τὸ Γ, οὕτως τὸ ΕΘ πρὸς τὸ Ζ· δι᾽ ἴσου ἄρα ἐστὶν ὡς τὸ ΑΗ πρὸς τὸ Γ, οὕτως τὸ ΔΘ πρὸς τὸ Ζ.

Ἐὰν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔχῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔχῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἕξει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον· ὅπερ ἔδει δεῖξαι.

Proposition 24†

If a first (magnitude) has to a second the same ratio that third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, then the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and sixth (added together, have) to the fourth.



For let a first (magnitude) AB have the same ratio to a second C that a third DE (has) to a fourth F. And let a fifth (magnitude) BG also have the same ratio to the second C that a sixth EH (has) to the fourth F. I say that the first (magnitude) and the fifth, added together, AG, will also have the same ratio to the second C that the third (magnitude) and the sixth, (added together), DH, (has) to the fourth F.

For since as BG is to C, so EH (is) to F, thus, inversely, as C (is) to BG, so F (is) to EH [Prop. 5.7 corr.]. Therefore, since as AB is to C, so DE (is) to F, and as C (is) to BG, so F (is) to EH, thus, via equality, as AB is to BG, so DE (is) to EH [Prop. 5.22]. And since separated magnitudes are proportional then they will also be proportional (when) composed [Prop. 5.18]. Thus, as AG is to GB, so DH (is) to HE. And, also, as BG is to C, so EH (is) to F. Thus, via equality, as AG is to C, so DH (is) to F [Prop. 5.22].

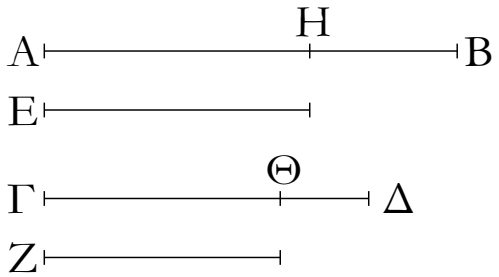
Thus, if a first (magnitude) has to a second the same ratio that a third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, then the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and the sixth (added

together, have) to the fourth. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\epsilon : \beta :: \zeta : \delta$ then $\alpha + \epsilon : \beta :: \gamma + \zeta : \delta$.

κε′.

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ μέγιστον [αὐτῶν] καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν.



Ἔστω τέσσαρα μεγέθη ἀνάλογον τὰ ΑΒ, ΓΔ, Ε, Ζ, ὡς τὸ ΑΒ πρὸς τὸ ΓΔ, οὕτως τὸ Ε πρὸς τὸ Ζ, ἔστω δὲ μέγιστον μὲν αὐτῶν τὸ ΑΒ, ἐλάχιστον δὲ τὸ Ζ· λέγω, ὅτι τὰ ΑΒ, Ζ τῶν ΓΔ, Ε μείζονά ἐστιν.

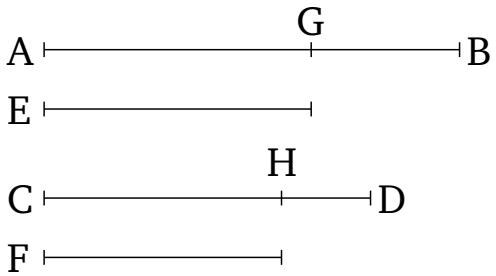
Κείσθω γὰρ τῷ μὲν Ε ἴσον τὸ ΑΗ, τῷ δὲ Ζ ἴσον τὸ ΓΘ.

Ἐπεὶ [οὖν] ἐστιν ὡς τὸ ΑΒ πρὸς τὸ ΓΔ, οὕτως τὸ Ε πρὸς τὸ Ζ, ἴσον δὲ τὸ μὲν Ε τῷ ΑΗ, τὸ δὲ Ζ τῷ ΓΘ, ἔστιν ἄρα ὡς τὸ ΑΒ πρὸς τὸ ΓΔ, οὕτως τὸ ΑΗ πρὸς τὸ ΓΘ. καὶ ἐπεί ἐστιν ὡς ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ, οὕτως ἀφαιρεθὲν τὸ ΑΗ πρὸς ἀφαιρεθὲν τὸ ΓΘ, καὶ λοιπὸν ἄρα τὸ ΗΒ πρὸς λοιπὸν τὸ ΘΔ ἔσται ὡς ὅλον τὸ ΑΒ πρὸς ὅλον τὸ ΓΔ. μεῖζον δὲ τὸ ΑΒ τοῦ ΓΔ· μεῖζον ἄρα καὶ τὸ ΗΒ τοῦ ΘΔ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν ΑΗ τῷ Ε, τὸ δὲ ΓΘ τῷ Ζ, τὰ ἄρα ΑΗ, Ζ ἴσα ἐστὶ τοῖς ΓΘ, Ε. καὶ [ἐπεὶ] ἐὰν [ἀνίσοις ἴσα προστεθῇ, τὰ ὅλα ἄνισά ἐστιν, ἐὰν ἄρα] τῶν ΗΒ, ΘΔ ἀνίσων ὄντων καὶ μείζονος τοῦ ΗΒ τῷ μὲν ΗΒ προστεθῇ τὰ ΑΗ, Ζ, τῷ δὲ ΘΔ προστεθῇ τὰ ΓΘ, Ε, συνάγεται τὰ ΑΒ, Ζ μείζονα τῶν ΓΔ, Ε.

Ἐὰν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ μέγιστον αὐτῶν καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν. ὅπερ ἔδει δεῖξαι.

Proposition 25†

If four magnitudes are proportional then the (sum of the) largest and the smallest [of them] is greater than the (sum of the) remaining two (magnitudes).



Let AB, CD, E, and F be four proportional magnitudes, (such that) as AB (is) to CD, so E (is) to F. And let AB be the greatest of them, and F the least. I say that AB and F is greater than CD and E.

For let AG be made equal to E, and CH equal to F.

[In fact,] since as AB is to CD, so E (is) to F, and E (is) equal to AG, and F to CH, thus as AB is to CD, so AG (is) to CH. And since the whole AB is to the whole CD as the (part) taken away AG (is) to the (part) taken away CH, thus the remainder GB will also be to the remainder HD as the whole AB (is) to the whole CD [Prop. 5.19]. And AB (is) greater than CD. Thus, GB (is) also greater than HD. And since AG is equal to E, and CH to F, thus AG and F is equal to CH and E. And [since] if [equal (magnitudes) are added to unequal (magnitudes) then the wholes are unequal, thus if] AG and F are added to GB, and CH and E to HD—GB and HD being unequal, and GB greater—it is inferred that AB and F (is) greater than CD and E.

Thus, if four magnitudes are proportional then the (sum of the) largest and the smallest of them is greater than the (sum of the) remaining two (magnitudes). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$, and α is the greatest and δ the least, then $\alpha + \delta > \beta + \gamma$.